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GROSS JOB CREATION, GROSS JOB DESTRUCTION AND EMPLOYMENT REALLOCATION*

by

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Abstract

This paper measures the heterogeneity of establishment-level employment changes in the U.S. manufacturing sector over the 1972 to 1986 period. Our empirical work exploits a rich data set with approximately 860,000 annual observations on 160,000 manufacturing establishments to calculate rates of gross job creation, gross job destruction, and their sum, gross job reallocation. The central empirical findings are as follows: (1) Based on March-to-March establishment-level employment changes, gross job reallocation averages more than 20% of employment per year. (2) For the manufacturing sector as a whole, March-to-March gross job reallocation varies over time from 17% to 23% of employment per year. (3) Time variation in gross job reallocation is countercyclic-gross job reallocation rates covary negatively with own-sector and manufacturing net employment growth rates. (4) Virtually all of the time variation in gross job reallocation is accounted for by idiosyncratic effects on the establishment growth rate density. Changes in the shape and location of the growth rate density due to aggregate-year effects and sector-year effects cannot explain the observed variation in gross job reallocation. (5) The part of gross job reallocation attributable to idiosyncratic effects fluctuates countercyclically. Combining (3)-(5), we conclude that the intensity of shifts in the pattern of employment opportunities across establishments exhibits significant countercyclic variation.

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I. Introduction and Overview

A. Objectives and Data

This paper measures the heterogeneity of establishment-level employment changes in the U.S. manufacturing sector over the 1972 to 1986 period. We document the magnitude and character of this heterogeneity, and we investigate patterns of cross-industry and time-series variation. We stress the implications of this heterogeneity for the turnover of jobs and workers, for labor market dynamics and for the nature of aggregate fluctuations.

Our empirical work exploits a tremendously rich data set with approximately 860,000 annual observations on 160,000 manufacturing establishments. The data are longitudinal and include observations on all manufacturing establishments sampled in the Annual Survey of Manufactures between 1972 and 1986. The combination of establishment-level longitudinal data, high frequency observations, a fifteen-year sample, and comprehensive coverage of the manufacturing sector provides an excellent basis for exploring the connection between the heterogeneity of establishment-level employment changes and aggregate fluctuations.

A key advantage of longitudinal establishment data for this purpose is that we are able to measure gross job creation, gross job destruction, and their sum, gross job reallocation. We calculate gross job creation as the sum of employment increases in new and expanding establishments. Similarly, we calculate gross job destruction as the sum of employment decreases in shrinking and dying establishments. Gross job reallocation represents the gross change in the number of employment positions at establishments.

Using the Longitudinal Research Datafile (LRD), we document (a) the magnitude of gross job creation, destruction and reallocation; (b) the cross-industry variation in average rates of gross job creation, destruction and reallocation; (c) the time variation in these measures for the manufacturing sector and for individual two-digit industries; (d) the contribution of (cross-sectional) mean aggregate, industry, region, and establishment size-class effects to the time variation in gross job creation, destruction and reallocation; and (e) the discreteness of establishment-level employment changes as reflected in, for example, the large fraction of job creation and destruction accounted for by establishment births and deaths. Our empirical work also quantifies the extent to which familiar characterizations of aggregate and industry-level fluctuations can account for fluctuations in gross job creation, destruction and reallocation.

B. Basic Facts About Gross Job Creation, Destruction and Reallocation

Before turning to the motivations for our empirical work and its implications, we review our main findings. From March 1972 to March 1986 (excluding the years 1974,

1979, and 1984 for reasons explained below), manufacturing employment contracted at an average net rate of 2.1% per year. Over the same period, manufacturing industries experienced high rates of gross job creation and destruction. Based on March-to-March establishment-level employment changes, we calculate that manufacturing's rates of gross job creation and destruction averaged 9.2% and 11.3% per year, respectively. Thus, one basic fact we uncover is that the heterogeneity of establishment-level employment changes implies large rates of gross job creation and destruction and, hence, large worker flows consequent to the reallocation of jobs across establishments. This fact has been documented before, perhaps most convincingly at high frequencies by Leonard (1987) and at low frequencies by Dunne, Roberts, and Samuelson (1989).

A second basic fact that emerges clearly from our study is the importance of establishment births and deaths in the process of job creation and destruction. Establishment deaths account for 25% of annual gross job destruction over the sample period, while establishment births account for 20% of annual gross job creation. Establishment deaths (births) occasionally account for one-third or more of gross job destruction (creation) during individual years. As a more general point, establishment-level employment changes exhibit considerable discreteness.

A third basic fact is that the average annual rate of gross job reallocation (the sum of the creation and destruction rates) varies greatly across two-digit manufacturing industries. Again based on March-to-March establishment-level employment changes, gross job reallocation rates range from 14% per year in both Tobacco and Paper Products to 27% per year in Apparel and 29% per year in Lumber and Wood Products. Looking across industries, there is a strong positive relationship between the time-series variability of net industry employment growth rates and average gross job reallocation rates within industries. In other words, industries that exhibit more volatile employment growth also exhibit greater heterogeneity of establishment-level employment fluctuations.

We document these basic facts about the magnitude, character, and cross-industry variation of job creation, destruction and reallocation in sections IV and V of the paper. Section II of the paper describes the LRD in detail, and section III explains the construction of our measures. In sections VI and VII we describe and implement a methodology for characterizing the nature of time-series variation in gross job creation, destruction and reallocation rates.

Two basic facts about time-series variation in gross job reallocation provide the stepping-off point for our analysis in sections VI and VII. As the first basic time-series fact, the job reallocation rate exhibits signficant time variation. The March-to-March job reallocation rate for the manufacturing sector ranges from a low of 17% in 1980 to a high of 23% in 1975 and 1983. For individual two-digit manufacturing industries, the range of

variation is typically much larger. For example, the March-to-March job reallocation rate in Primary Metals varies from 13% to 30% per year.

As the second basic time-series fact, time variation in the job reallocation rate is consistently countercyclic. All twenty of the two-digit manufacturing industries show a negative time-series correlation between the rates of own net employment growth and gross job reallocation. The mean correlation is -0.5. In addition, nineteen out of twenty industries exhibit a negative correlation between total manufacturing's rate of employment growth and the industry job reallocation rate.

C. Empirical Methodology

These two basic facts about time variation in gross job reallocation immediately prompt several important and related questions: What factors account for the countercyclic time variation in gross job reallocation? Is this countercyclic time variation driven largely by aggregate, sectoral or idiosyncratic effects? Does the countercyclic variation in gross job reallocation simply reflect familiar patterns of differential sectoral responses to business cycle fluctuations? We develop and implement an empirical methodology for addressing these questions.

Our methodology relies on simple models of establishment-level employment growth to quantify the contribution of aggregate, sectoral and idiosyncratic effects to changes in the location, spread and shape of the empirical density function over establishment growth rates. Our measures of gross job creation, destruction and reallocation can be interpreted as the outcome of integrating over the empirical density function of establishment growth rates with appropriate integrands. Thus, we first specify an establishment growth rate model in terms of aggregate-year effects, sector-year effects, and time-varying idiosyncratic effects. Second, we apply the growth rate model to isolate the contributions of aggregate, sectoral and idiosyncratic effects to changes in the shape and location of the empirical density function. Third, we recompute the time series on gross job creation, destruction and reallocation by integrating over the time series of appropriately adjusted empirical density functions. Fourth, using the original and adjusted time series, we apply simple variance decompositions to quantify the contribution of aggregate, sectoral and idiosyncratic effects to the time variation in gross job creation, destruction and reallocation. Finally, we examine the time-series behavior of that part of gross job reallocation driven by idiosyncratic effects.

D. The Major Empirical Result of the Paper

The results of applying our methodology are striking and consistent. The overwhelming bulk of the time variation in gross job reallocation is accounted for by time variation

in the idiosyncratic component. Aggregate-year effects and sector-year effects account for a small fraction of time variation in gross job reallocation. Furthermore, the idiosyncratic contribution to the gross job reallocation rate exhibits a strong pattern of countercyclic movements with respect to own-sector and total manufacturing net employment growth rates. These results hold for every sectoral classification scheme we considered: two-digit industry, four-digit industry, geographic region, two-digit industry and geographic region simultaneously, and establishment size class. Thus, the data point to a systematic negative relation between net aggregate or sectoral employment growth and the heterogeneity of establishment-level employment changes.

It is useful to recast these results in terms of time variation in gross job creation and destruction rates. Note, first, that aggregate-year effects represent simple mean translations of the size-weighted empirical density function over establishment growth rates. In the data, aggregate-year effects typically shift much of the density's mass from one side of zero to the other and thereby account for large, and roughly offsetting movements, in gross job creation and destruction. In this respect, our model specification and empirical results are consistent with traditional and widely-held views about the nature of aggregate fluctuations. But the results described in the previous paragraph indicate that this view of aggregate fluctuations–even when suitably modified to accomodate differential sectoral responses to business cycles–is seriously incomplete. The results indicate, for example, that the rise in gross job destruction that accompanies aggregate economic contraction cannot be fully accounted for by aggregate-year effects and sector-year effects. While these effects play a major role, they are strongly reinforced in their impact on job destruction by idiosyncratic effects. Gross job destruction rises during aggregate contractions both because of a leftward shift *and* because of an increased spread in the empirical density function. Likewise, the fall in gross job creation that accompanies aggregate economic contraction is significantly dampened by an increased spread in the empirical density.

This paper's major empirical result can be succinctly characterized as follows. Consider the pattern of employment opportunities across establishments. Quite obviously, the pattern of employment opportunities shifts over time. Less obviously, but emerging clearly from this and other studies, the pace or intensity of these shifts is of impressive magnitude. Finally comes the question of time variation in the intensity of shifts in the pattern of oppurtunities. Our work shows that the intensity of shifts in the pattern of employment opportunities across establishments exhibits significant countercyclic variation.

E. Relationship to the Literature and Implications for Macroeconomics

The idea that continuous shifts in the pattern of employment opportunities cause significant worker reallocation and a nontrivial amount of unemployment began to take

firm root among economists in the 1960's, as evidenced by the work of Solow (1964), Holt and David (1966), Friedman (1968), and Phelps (1968). The idea has since been formalized in numerous papers that develop the implications of search, matching, and specific human capital investment technologies for theories of unemployment and labor market dynamics. Lucas and Prescott (1974) and Hall (1979) represent well-known early examples. More recently, Pissarides (1986) and Blanchard and Diamond (1989) investigate the joint dynamics of vacancies and unemployment within frameworks that successfully maintain a tight link between theory and empirics. We interpret our evidence on the magnitude and time variation of gross job reallocation as providing a strong impetus to further research on aggregate fluctuations that explicitly treats the gross flows of workers and jobs and the connection of these flows to unemployment, vacancy and wage dynamics.[1]

In a closely related strand of the literature, a controversy has arisen over the role that changes in the intensity of shifts in the pattern of employment opportunities play in aggregate economic fluctuations. This controversy dates to the publication of a provocative paper by Lilien (1982a) on the contribution of inter-industry shifts in labor demand to aggregate unemployment fluctuations. Lilien found a strong positive relationship in time-series data between unemployment and a measure of the cross-industry dispersion in net employment growth rates. In his interpretation of this finding, Lilien (1982a) argued that fluctuations in the volatility of labor demand induced substantial fluctuations in labor reallocation and unemployment in the postwar U.S. economy. His argument and interpretation of empirical results explicitly appealed to the time-consuming nature of worker reallocation and, consequently, the potential for sectoral disturbances to drive aggregate fluctuations. Lilien's paper inspired a rich variety of responses and has been followed by much related research.[2]

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[1] In section IV, we use our measures of gross job creation and destruction to compute bounds on the number of workers who change jobs or labor force status between one March and the next as a direct result of job reallocation. Over the sample period, the lower and upper bounds on the number of workers involved in this reallocation process average 13% and 21% of manufacturing employment per year. Thus, our results show that a large amount of worker mobility is demand-driven in the sense of representing a direct response to the reallocation of jobs across establishments.

[2] See Darby, Haltiwanger and Plant (1985,1987), Abraham and Katz (1986), Hall and Lilien (1986), Loungani (1986), Davis (1985, 1987a,b), Leonard (1987, 1988), Murphy and Topel (1987), Neelin (1987), Samson (1987), Akerlof, Rose and Yellen (1988), Holzer (1988), Katz (1988), Lilien (1982b,1988), Mork (1989), Mork, Mysen and Olsen (1988), Neumann and Topel (1988), Topel and Weiss (1988), Blanchard and Diamond (1989),

Abraham and Katz (1986) challenged Lilien's interpretation of his empirical finding. They showed that a positive time-series relationship between unemployment and Lilien's cross-industry dispersion measure could entirely reflect differences in the timing and magnitude of sectoral employment growth rate responses to aggregate-demand driven business cycle fluctuations. Recognizing this problem, Lilien (1982b), Davis (1985), Abraham and Katz, and several other researchers constructed alternative proxies for the intensity of inter-industry demand shifts. Essentially, these alternative proxies represented attempts to construct dispersion measures from industry employment growth rates that were purged of the effects of aggregate demand disturbances. This approach failed to resolve the issue of whether sectoral shifts in labor demand drive a large fraction of unemployment fluctuations, in part because empirical results turned out to be sensitive to the choice among alternative plausible specifications.[3]

Our measure of gross job reallocation, adjusted to remove the contributions of aggregate-year effects and sector-year effects, is immune to the criticism that Abraham and Katz directed toward Lilien's (1982a) measure. In particular, our adjusted job reallocation measure allows for completely unrestricted mean sectoral responses to aggregate disturbances. We place neither linearity, magnitude, nor timing restrictions on these responses. Our only restrictions are those inherent in the sectoral classification scheme, and we make a serious effort to consider several classification schemes that are commonplace in business cycle analyses. Our ability to circumvent the problems stressed by Abraham and Katz stems directly from our use of longitudinal establishment-level data.

As we noted above, our adjusted gross job reallocation measures exhibit significant countercyclic time variation. In this respect, our major finding supports Lilien's interpretation of his empirical results. However, two caveats should serve as a caution to the

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Loungani and Rogerson (1989), and Loungani, Rush and Tave (1989). Purely theoretical research on the aggregate implications of sectoral disturbances includes papers by Black (1987), Hamilton (1987), Hosios (1988), Riordan and Staiger (1988), Williamson (1988). and Rogerson (1989).

[3] Abraham and Katz suggested another approach to resolving the issue of how to interpret Lilien's (1982a) empirical finding. They argued that the negative relationship between the unemployment rate and the vacancy rate in time-series data supports the aggregate demand interpretation of Lilien's finding. Blanchard and Diamond (1989) rely on a related identifying assumption to conclude that "rellocation shocks" play little role in high frequency unemployment fluctuations. In our view, unemployment-vacancy correlations are unable to distinguish between the alternative interpretations of Lilien's finding. See Davis (1987b) and Hosios (1988) for discussion on this point.

reader not to conclude too much from this statement. First, we make no attempt to investigate the relationship between unemployment and adjusted gross job reallocation in this paper. (We develop an empirical framework for this purpose in Davis and Haltiwanger (1988).) Second, and more fundamentally, resolution of the problem stressed by Abraham and Katz is not sufficient to settle the central issue in the sectoral shifts literature. The central issue is whether (a) time variation in the intensity of disturbances to the structure of labor demand drives aggregate fluctuations, or (b) aggregate demand disturbances drive the time variation in the pace of job reallocation. In other words, the central issue turns on the question of causal direction in the relationship between aggregate fluctuations and the pace of job reallocation.

To appreciate that our empirical results are consistent with a causal arrow running in either or both directions, consider three plausible scenarios. One scenario stresses the interaction between vintage capital effects, location choice, production technique, etc. and cost shocks, e.g., energy price shocks. When these sources of heterogeneity among establishments are important, energy price shocks cause sharply different establishment-level employment changes, even among establishments in narrowly defined sectors. A second scenario stresses product market heterogeneity in the face of a terms of trade (or energy) shock. Some establishments produce goods that are highly substitutable with imports, while other establishments in the same sector produce goods that are poorly substitutable with imports. A terms of trade disturbance causes sharply different establishment-level employment changes, again even among establishments within the same sector. (Provided that workers have sector specific attachments, employment can rise in some establishments and fall in others within the same sector.) Given the importance of frictions in the labor market, the increased job reallocation caused by energy and trade shocks in these two scenarios leads to fluctuations in aggregate activity. Thus, in both scenarios the causal arrow runs from disturbances to the structure of labor demand to aggregate fluctuations. Analyses by Black (1987), Hamilton (1987), Hosios (1988), and Rogerson (1989) capture important aspects of these scenarios.

A third scenario posits an economy subject to aggregate demand disturbances and characterized by a time-invariant variance of idiosyncratic establishment-level disturbances. With respect to technologies, this scenario posits that workers and firms face opportunity costs of foregone production associated with mobility and expansion or contraction. For workers, foregone production can arise from the time costs of search, matching, mobility, or learning by doing. For firms, foregone production can arise from costs of adjusting the labor input. In this scenario, anticipated future movements in aggregate demand imply incentives to intertemporally substitute the reallocation of jobs and workers from periods of high opportunity costs to periods of low opportunity costs. Since

aggregate demand disturbances affect the opportunity cost of foregone production, they drive the pace of reallocation. This reallocation timing effect is modelled in Davis (1985) and discussed in Darby, Haltiwanger and Plant (1985), Davis (1987b), and Loungani and Rogerson (1989).

II. The LRD and other Longitudinal Establishment-Level Data Bases

Despite the magnitude of labor reallocation implied by establishment-level employment dynamics, and the centrality of the reallocation process in many theories of unemployment, the role of establishment-level labor demand disturbances in the secular and cyclical behavior of the labor market remains largely unexplored. The paucity of work in this area primarily reflects data limitations. In this section, we discuss recent work that investigates the magnitude of worker and/or employment reallocation across establishments or firms, and we discuss the limitations of the data on which this work is based. Beyond serving as a brief literature review, the discussion highlights the advantages of the LRD.

Two papers by Leonard (1987, 1988) document high rates of gross job creation and destruction relative to net employment changes. Leonard (1987) uses annual data on the population of establishments participating in the Wisconsin unemployment insurance program during the years 1977-82. Besides being limited to a single state for a five-year period, the Wisconsin data suffer from two other shortcomings. First, it is difficult to longitudinally link establishments before and after ownership transfers. Second, depending on the reporting method chosen by the firm, it is often impossible to obtain employment observations on the separate establishments of multi-establishment firms. In more recent work, Leonard (1988) constructs a longitudinal sample of annual establishment-level employment observations using firm responses to federal EEO-1 forms for the years 1978 to 1984. Leonard exploits these data to investigate the relationship between measures of gross job reallocation and unemployment across SMSAs. The primary drawbacks of this data set are the severe undersampling of small and medium-sized establishments and firms, and the systematic exclusion of births, deaths, and firms with volatile employment histories. Thus, Leonard's (1988) sample involves implicit selection along a number of dimensions likely to be correlated with patterns of gross job creation and destruction.

Another longitudinal database that has been used to measure gross job reallocation and related concepts is the U.S. Establishment and Enterprise Microdata file (USEEM) put together by the Small Business Adminstration from Dun and Bradstreet data.[4] Several studies (see, e.g., Birch (1981), Armington and Odle (1982) and Howland (1988)) use these data to calculate job destruction and creation rates over two or three year intervals from firm-level employment changes. All of these studies report high rates of gross job creation and destruction relative to net changes. However, there are several problems with the USEEM. First, since it is limited to firms that purchase or sell on credit, the

[4] Our discussion of the USEEM draws on the useful and more detailed discussions in Evans (1987), Howland (1988), and *The State of Small Business, 1987.*

USEEM undersamples small and younger firms. Second, the USEEM lacks a contemporaneous employment observation (corresponding to the observation year) for many firms. Third, the USEEM lacks information on mergers and acquisitions, precluding a careful distinction between ownership transfers (partial or complete) and true births, expansions, contractions, and deaths. Fourth, as Howland's (1988) analysis indicates, entry into and exit from the Dun and Bradstreet data correspond very imperfectly to birth and death. Fifth, the sheer size of the USEEM–it contains data on a few million firms–inhibits careful consistency checks and data cleaning.

Finally, we note that Dunne, Roberts and Samuelson (1987,1989) use the Census of Manufactures portion of the LRD to compute gross job creation and destruction rates over five-year intervals from 1963 to 1982. Their research documents the tremendous heterogenity in establishment-level employment changes at low frequencies. Unfortunately, the five-year intervals between Censuses preclude an analysis of gross job creation and destruction at business cycle frequencies.

Our study exploits the annual data in the LRD. In addition to containing high-frequency observations, the LRD overcomes the potential selection and mismeasurement problems identified above by including almost all manufacturing employment within its sampling frame and by treating establishments, rather than financial entities, as the data unit. An establishment is defined as a single physical location engaged in manufacturing activity. The only manufacturing establishments excluded from the sampling frame of the LRD are those with fewer than five employees. These establishments account for only one percent of manufacturing employment, based on tabulations from either the Census of Manufactures or County Business Patterns.

The LRD is basically a series of contiguous five-year panels with annual and some quarterly data on manufacturing establishments, plus Census-year data on the universe of manufacturing establishments with more than five employees. Census years in the LRD are 1963, 1967, 1972, 1977, and 1982. From the Census-year universe, the Bureau draws a sample of establishments which are then surveyed during five successive years. This five-year panel, which commences two years after a Census year, comprises the sample of establishments that makes up the Annual Survey of Manufactures (ASM). New establishments are added to the panel as it ages to incorporate births and to preserve the representative character of the panel. In 1977, the LRD included roughly 70,000 out of the 360,000 establishments in manufacturing industries.

Variables in the LRD include plant location (street address, county, state); detailed SIC code; coverage code (responses include no change in operations, plant dismantled, plant idle but still owned, newly opened operation, previously idle establishment that resumed activity, etc.); sample weight of establishment; total employment during the payroll

period covering the 12th of March; a permanent plant identification (PPN) number; and other establishment and firm identification numbers that have been used by the Bureau for various reasons. The PPN number allows us to construct longitudinal data for individual establishments.[5]

Four other aspects of the LRD merit careful consideration and discussion: (i) the sampling procedures used to create the panels, (ii) the assignment of establishment weights for aggregation purposes, (iii) the treatment of births and deaths, and (iv) the relationship between our estimates of employment changes and published ASM totals. We take these matters up in turn.

With respect to the five-year ASM panels, establishments fall into three broad groups. As noted, the group containing establishments with fewer than five employees is excluded from the sampling frame. A second group of establishments is included in the panel with certainty. For the 1979-83 panel, for example, the certainty group includes all establishments with 250 or more employees during the 1977 Census year. This certainty threshhold is lower in some industries, and many establishments are included with certainty based on other criteria. Taken as a whole, the certainty cases account for about two-thirds of manufacturing employment in the 1979-83 panel. Establishments that fall into neither of the first two groups are sampled with probabilities proportional to a measure of size determined for each establishment from the preceding Census. Sampling probabilities for non-certainty establishments range from 1.000 to 0.005. Some, but not most, of the non-certainty establishments appear in contiguous panels. Thus, our ability to link establishment-level observations across panels ranges from excellent for large establishments to quite poor for the smallest establishments. Since the industry-level measures that we construct from the LRD are computed from establishment-level employment changes, the quality these measures is lower in the first period of each panel. While the available data provide considerable scope for estimating industry-level gross changes during the first period of each panel, we opted for the simpler procedure of deleting those periods from our sample.

The Bureau assigns to each establishment a sample weight equal to the reciprocal of its probability of selection into the panel. The sample weights are available to us

[5] In constructing the longitudinal data files, we discovered (with much assistance from Tim Dunne and Cyr Linonis) approximately 5000 cases of misassigned PPN numbers. We have implemented procedures that exploit Census Tab numbers, as well as name and location information, to correct the misassigned PPN numbers.

and we use them in constructing the measures detailed below.[6] The sample weight for an establishment typically remains fixed throughout a five-year panel, although sample weights occasionally change due to ownership transfer or a change in industry.

For our purposes, the most serious data quality issue in the LRD is the treatment of establishment births. The manner in which establishment births are incorporated into ASM panels differs for single and multi-unit companies. First, consider births of single-unit companies. Each year the Census Bureau obtains a list of new establishments-actually, assignments of new social security employer identification (EI) numbers–from the Social Security Administration. The Bureau takes this list of potential births and attempts to verify the true births. Historically, about half of these new employer identification numbers represent true births, while the other half represent ownership transfers or some other restructuring of existing businesses. If the Bureau determines that a new EI number represents a true birth and the employment at the establishment is greater than 35, it is included in the panel with a sample weight of one. Small single-unit births are added to the aggregate totals through an impute block. Births to multi-unit companies are detected through the annual Company Organization Survey. A sample of these multi-unit births is drawn and added to the panel with appropriately calculated sample weights.

One key issue is whether the timing of births is accurately measured. Due primarily to the verification processing time, it appears that births are often incorporated into the panel the year *after* true birth. However, since we analyze March-to-March employment changes, this lag poses relatively minor measurement problems. As long as a birth occurs after March 12, our dating of the gross job creation associated with that birth will be accurate.

Another key issue is whether births and deaths are accurately detected. Accurate measurement of births depends on the reliability of processing the new EI numbers and the accuracy of the corresponding coverage code variable. Similarly, accurate measurement of deaths depends critically on the reliability of the coverage code variable. In this paper, we adopt a conservative stance regarding the idenfication of births and deaths. In particular, we do not use entry or exit from the panel (as in some other studies) as an indicator of birth or death; rather, we rely on the coverage code.

A final data issue involves comparisons between our figures and published ASM employment figures. Even though we use the ASM sample and the appropriate sample

[6]The Bureau's sampling scheme is designed to optimize estimates of the *level* of industry activity, while our measures are based on *changes* in establishment-level employment. Nonetheless, given the sample sizes of the ASM panels and the aggregation used to construct our measures, sampling error is a minor concern in this study.

weights, discrepancies arise between the net employment change estimates we calculate and the net employment change estimates implied by the published ASM totals. There are several sources of this discrepancy, most of which are associated with definitional differences.

For example, excluding 1974, 1979, and 1984 (because of cross-panel linkage problems) the 1972-1986 average net employment growth calculated in a standard manner from published ASM totals is -.8% per year. This compares to our estimate of -2.1% per year. This discrepancy can be decomposed as follows. First, published ASM totals are annual averages of four quarterly employment observations. In contrast, we use March-to-March changes. This difference accounts for approximately .5 percentage points of the discrepancy. Second, our estimate of -2.1% uses the average of the current year and the previous year employment as the base for computing rates (see section III below for motivation of our measure). This difference accounts for approximately .2 percentage points of the discrepancy. Third, the published ASM aggregates are not simply weighted sums of establishment data in the ASM. Instead, the published data reflect a set of ex post adjustments to the weighted sample totals. These ex post adjustments account for approximately .5 percentage points of the discrepancy. The remaining discrepancy (approximately .1 percentage points) reflects our use of coverage code information, rather than sample entry and exit, to measure births and deaths. For a discussion of Census procedures for constructing the published ASM figures and a full accounting of discrepancies between weighted-ASM and published figures, see Davis, Haltiwanger, and Schuh (1989).

III. Gross Job Creation, Gross Job Destruction, and Related Measures

A. Employment Size and Growth Rate Measures

Our empirical work below investigates cross-industry and time variation in the empirical density function over establishment growth rates. In large part, our objective is to characterize the relationship between changes in the central location of the density and changes in its spread or shape. In making this characterization, it is important to integrate the treatment of births and deaths into the analysis of expanding and contracting establishments. Unfortunately, the unboundedness and asymmetry of conventional growth rate measures hamper an integrated treatment of births, expansions, contractions, and deaths.

These considerations motivate our use of an alternative growth rate measure defined on a base that equals the simple average of current and lagged employment. The resulting growth rate measure is bounded and symmetric about zero. It is a straightforward monotonic transformation of the conventional growth rate measure. Furthermore, as will

become apparent, our alternative measure facilitates an integrated, symmetric treatment of births and deaths in the empirical analysis. We now introduce some notation and formally define our growth rate measure.

Let

$$EMP_{et} \quad \text{for} \quad e = 1, 2, \ldots, E_t; \quad t = 1, 2, \ldots,$$

denote time-t employment at establishment e. E_t denotes the number or set of establishments in the LRD at time t. To accomodate births, we include establishments born at time $t+1$ in the set E_t with EMP_{et} equal to zero by construction.

Let EMP_{it} be total employment in industry i at time t. EMP_{it} satisfies

$$EMP_{it} = \sum_{e \in E_{it}} w_{et} EMP_{et},$$

where E_{it} denotes the set of establishments in industry i at time t, and where w_{et} is the sample weight applied to establishment e, equal to the reciprocal of its sampling probability. Other employment aggregates introduced below are similarly constructed as a weighted average of LRD-sampled establishments. We suppress the sample weights in the exposition that follows, although they are applied in the actual construction of the measures.

As our measure of establishment size at time t, we use $x_{et} = .5(EMP_{et} + EMP_{e,t-1})$. Similarly, $X_{it} = .5(EMP_{it} + EMP_{i,t-1})$ defines the size of industry i at time t, and $X_t = .5(EMP_t + EMP_{t-1})$ defines the size of the manufacturing sector at time t.

Using this concept of size,

$$g_{et} = \frac{\Delta EMP_{et}}{x_{et}}$$

defines the time-t growth rate for establishment e. Industry and manufacturing employment growth rates, g_{it} and g_t, are defined analogously. These growth rate measures lie in the closed interval $[-2, 2]$ with deaths (births) corresponding to the left (right) endpoint. g_{et} is monotonically related to the conventional growth rate measure:

$$G_{et} \equiv \frac{\Delta EMP_{et}}{EMP_{e,t-1}} = \frac{2g_{et}}{2 - g_{et}}.$$

As this relationship indicates, g_{et} approximately equals G_{et} for small values of g_{et}. The approximation is accurate for the growth rate values typically assumed by industry employment aggregates in annual and quarterly data. Hence, the use of X_{it} rather than EMP_{it} has little effect on the calculated values of industry-level rates of net job creation,

gross job creation, etc. When appropriate, we express our empirical results in terms of both g_{et} and G_{et}.

B. Gross Job Creation and Destruction Measures

To construct an industry-level measure of gross job creation, we simply add up employment growth at expanding and new establishments within the industry. Similarly, our measure of gross job destruction simply sums employment losses over shrinking and dying establishments within the industry. To express these measures as rates, we divide by our measure of industry size. Thus,

$$POS_{it} = \frac{1}{2} \sum_{e \in E_{it}} \frac{(|\Delta EMP_{et}| + \Delta EMP_{et})}{X_{it}} = \frac{1}{2} \sum_{e \in E_{it}} (\frac{x_{et}}{X_{it}})(|g_{et}| + g_{et}) \qquad \text{and}$$

$$NEG_{it} = \frac{1}{2} \sum_{e \in E_{it}} \frac{(|\Delta EMP_{et}| - \Delta EMP_{et})}{X_{it}} = \frac{1}{2} \sum_{e \in E_{it}} (\frac{x_{et}}{X_{it}})(|g_{et}| - g_{et})$$

define gross job creation and destruction rates in industry i at time t.

We make three straightforward but important observations about these measures. First, it seems apparent that idiosyncratic changes in establishment-level employment are driven by changes in desired establishment size rather than by changes in the stock of unfilled positions, especially for sampling intervals of one year. For this reason, POS_{it} and NEG_{it} reflect the reallocation of employment positions or jobs, and not the reallocation of workers. Of course, one motivation for our research is that the reallocation of jobs partly drives the reallocation of workers. Thus, the job reallocation concept in this paper differs from, but is related to, the worker turnover concepts considered by Lilien (1980), Hall (1982), Akerlof, Rose and Yellen (1988), and others.

Second, since the LRD contains point-in-time employment observations, transitory contractions and expansions are less likely to be captured by POS_{it} and NEG_{it} than longer-lived establishment-level employment changes.

Third, since we observe only plant-level employment, we cannot determine whether a given level of employment in two different periods for the same plant represents the same or different employment positions. This observation and the point-in-time nature of the employment data imply that POS_{it} and NEG_{it} represent lower bounds on gross job creation and destruction.

C. Gross Job Reallocation and Bounds on Worker Reallocation

We use the sum of POS_{it} and NEG_{it}, SUM_{it}, to measure the gross job reallocation rate in industry i between $t-1$ and t. $X_{it}SUM_{it}$ represents the gross change in the

number of employment positions at establishments. $X_{it}SUM_{it}$ also represents an upper bound on the number of workers who change jobs (or labor force status) in direct response to establishment-level employment changes in industry i. (The interpretation of $X_{it}SUM_{it}$ as an upper bound is subject to the qualifications about the lower-bound nature of POS_{it} and NEG_{it} discussed in the preceding section.) $X_{it}SUM_{it}$ represents an upper bound because some workers move from shrinking to growing establishments within industry i between $t-1$ and t. To obtain a lower bound, we eliminate the possibility of double counting job losers who move directly to new jobs at expanding establishments in the same industry. That is, $X_{it}MAX_{it} = X_{it}\text{Max}\{POS_{it}, NEG_{it}\}$ represents a lower bound on the number of workers who change jobs (or labor force status) in direct response to job reallocation in industry i. In line with this discussion, we often refer to SUM_{it} and MAX_{it} as upper and lower bounds on the rate of worker reallocation driven by job reallocation.

D. Partitions of Gross Job Creation and Destruction by Establishment Growth Rate

Gross job creation and destruction are distributed over establishments experiencing a range of expansion and contraction rates. To characterize the shape of this distribution, we partition gross job creation, for example, into intervals that answer the question: "What fraction of gross job creation is accounted for by establishments that expand at a rate greater than y and less than z?" Formally,

$$POS_{it}(y,z) = \frac{1}{2X_{it}POS_{it}} \sum_{\substack{e \in E_{it} \\ g_{et} \in (y,z]}} (\Delta|EMP_{et}| + \Delta EMP_{et}),$$

and similarly for $NEG_{it}(y,z)$, where $y, z \in [-2,2]$.

To the extent that slow-growing establishments account for the bulk of gross job creation, $POS_{it}(y,z)$ is large for small values of y and z, say $y = 0$ and $z = .1$. Conversely, if fast-growing establishments account for the bulk of gross job creation, then $POS_{it}(y,z)$ is large for large values of y and z, say $y = 1$ and $z = 1.5$. Thus, the $POS_{it}(y,z)$ and, similarly, $NEG_{it}(y,z)$ partitions describe the distribution of gross job creation and destruction across establishments experiencing differing growth rates. Note that our partitions of gross job creation and destruction are refinements of the usual distinction among births, expansions, contractions and deaths. See Dunne, Roberts and Samuelson (1989), for example.

IV. Job Creation, Destruction and Reallocation: Magnitude and Variation

A. Magnitude and Cross-Industry Variation

Table 1 presents annual average rates of gross job creation, gross job destruction, net employment growth, gross job reallocation, and a lower bound on worker reallocation for the manufacturing sector and for each two-digit industry. The industry figures are X_{it}-weighted averages of annual observations for the years 1973, 1975-1978, 1980-1983, and 1985-1986. The figures for the manufacturing sector are X_i-weighted averages of the industry figures.

The $\overline{NET}_{it}$ column of Table 1 shows that employment contracted in every two-digit manufacturing industry over the sample. Annual net contraction rates range from .2% in Instruments to 5.4% in Primary Metals. The manufacturing sector as a whole declined at a rate of 2.1% per year.

Despite pervasive and often sharp secular employment contractions in the manufacturing sector, every two-digit industry experienced significant gross job creation. Average March-to-March gross job creation rates range from 5.8% in Tobacco to 12.9% in Lumber and Wood Products. March-to-March gross job destruction rates range from 7.8% in Paper to 16.0% in Lumber and Wood Products. In the manufacturing sector as a whole, gross job creation and destruction rates averaged 9.2% and 11.3%, respectively.

The last two columns in Table 1 report rates for gross job reallocation and a lower bound on the worker reallocation driven by creation and destruction of jobs. These measures point to a tremendous reallocation of employment positions and workers in response to establishment-level employment changes. The annual average job reallocation rate ranges from 14.0% in Tobacco to 28.8% in Lumber and Wood Products. The average job reallocation rate for the manufacturing sector exceeds twenty percent per year. Recall that this figure represents an upper bound on the job-reallocation-driven rate of worker reallocation. The lower bound on the job-reallocation-driven rate of worker reallocation ranges from 8.9% in Chemicals and Paper to 18.8% in Lumber and Wood Products. For the manufacturing sector as a whole, the lower bound averages nearly thirteen percent per year.

Table 1 also reports size-weighted standard deviations of the industry averages. The cross-industry standard deviation of average gross job destruction rates equals 1.5%, which exceeds the cross-industry standard deviation of gross job creation rates by roughly forty per cent. The size-weighted cross-industry standard deviation of $\overline{NET}_{it}$ equals 1.1%, whereas the standard deviation of $\overline{SUM}_{it}$ equals 2.4%.

Given the variance decomposition implied by the identity, $NET \equiv POS - NEG$, these figures imply a strong positive covariance between industry average rates of gross job creation and industry average rates of gross job destruction. Indeed, the cross-industry correlation between $\overline{POS}_{it}$ and $\overline{NEG}_{it}$ is .764, which differs from zero at a marginal signigicance level of .0001 in a two-tailed test. Thus, industries characterized by high

average rates of gross job creation typically experienced high average rates of gross job destruction as well.

B. Time-Series Variation

Table 2 presents information on the time variation in the various measures. The table entries report the total manufacturing observations for t equal to 1973, 1975-1978, 1980-1983, and 1985-1986.

Manufacturing employment contracted during seven out of the eleven years in the sample. The most severe contraction occurred during 1975, when net and gross job destruction rates reached 10.0% and 16.6% of manufacturing employment. Net and gross job creation rates peaked in 1973 at 6.1% and 13.2% of employment.

The lower bound on the worker reallocation rate varies from a low of 10.2% in 1980 to a high of 16.6% in 1975. The job reallocation rate ranges from 17.3% in 1980 to 23.3% in 1975. We note here that the rates of net employment growth and gross job reallocation covary negatively–their time-series correlation coefficient is -.57, based on the observations in Table 2. Furthermore, all twenty two-digit industries show a negative time-series correlation between the rates of own net growth and own job reallocation. Thus, the data show a clear pattern of countercyclic variation in the job reallocaton rate.[7]

Table 3 reports the time-series standard deviations of NET_{it}, POS_{it}, NEG_{it}, and SUM_{it} for each two-digit industry and for the manufacturing sector. The volatility of net growth rates varies greatly across industries, as do the volatilities of gross job creation, destruction and reallocation rates. In terms of net growth, Lumber and Wood Products is the most volatile industry, showing a time-series standard deviation of 10.2% per year. At the other extreme, the standard deviation of the net growth rate in Food and Kindred Products is only 2.4% per year. In terms of gross job creation and destruction, the volatility measures for these two industries also lie at or near the extremes. In Food, the standard deviation equals 1.4% per year for job creation and 1.6% per year for job destruction. In Lumber, the standard deviation equals 4.8% per year for job creation and 5.9% per year for job destruction. Primary Metals shows the greatest volatility of gross job destruction rates with a standard deviation of 6.3% per year.

[7]Note that $\mathrm{COV}(NET_t, SUM_t) = \mathrm{V}(POS_t) - \mathrm{V}(NEG_t)$. Thus, a negative correlation between NET_t and SUM_t means that the time-series variability of NEG_t exceeds that of POS_t. In section VII below, we show that this empirical relationship is driven by the fact that gross job destruction increases more, and gross job creation fall less, during contractions than can be explained by aggregate-year and sector-year effects.

The standard deviation of the gross job reallocation rate ranges from 1.4% per year in Chemicals to 4.9% per year in Primary Metals. The standard deviation of the gross job reallocation rate for the manufacturing sector as a whole equals 1.6% per year, only 59% of the size-weighted average of the industry standard deviations. This comparison indicates that much of the time-series variation in industry-level job reallocation cannot be explained by factors that have the same effect across industries.

C. Summary

Three broad empirical regularities established in this section merit reiteration and emphasis. First, the March-to-March job reallocation rate in the manufacturing sector is impressively large–it averages nearly twenty percent per year. The number of workers who switch establishments or labor force status in direct response to establishment-level job reallocation is bounded between thirteen and twenty-one per cent of manufacturing employment in an average year. These bounds indicate that a substantial fraction of total worker reallocation is demand driven in the sense of representing a direct response to establishment-level employment changes. Second, average rates of gross job creation, destruction and reallocation vary by a factor of two or more across industries. Third, gross job reallocation varies countercyclically for the manufacturing sector as a whole and for every two-digit industry. For the manufacturing sector as a whole, the gross job reallocation rate varies from a low of 17.3% in 1980 to a high of 23.3% in 1975.

V. The Distribution of Establishment Growth Rates

A. The Size-Weighted and Unweighted Growth Rate Distributions

Figures 1A and 1B present histograms of all growth rate observations on manufacturing establishments in our sample. The histograms depict the shape of the empirical density over the 677,000 annual observations on g_{et}. In Figure 1A the observations on the g_{et} are weighted by size (x_{et}), while in Figure 1B the observations on the g_{et} are unweighted. The mean of the size-weighted distribution depicted in Figure 1A equals the annualized rate of contraction in the manufacturing sector over the sample period.

Both the weighted and unweighted growth rate densities are roughly symmetric with central peaks in the interval surrounding zero. The histogram peak in Figure 1A indicates that, on a size-weighted basis, 29% of the observations on annual establishment growth rates fall in the interval [-.05,.05]. Again on a size-weighted basis, about 63% of annual growth rate observations fall in the interval [-.15,.15].

Births and deaths account for only 2.4% of all size-weighted growth rate observations. Two caveats should be borne in mind when interpreting this aspect of the size-weighted

density. First, our size metric (x_{et}) assigns only half as much weight to observations on births and deaths as would a more conventional size metric. For example, if we were to ask what fraction of all current employment occurs at establishments that were born within the past year, the birth category would appear twice as important as in Figure 1A. Second, although births and deaths account for a small fraction of size-weighted establishment growth rate observations, they account for a large fraction of gross job reallocation. We return to this point below.

A visual comparison between Figures 1A and 1B reveals a striking difference between the size-weighted and unweighted distributions over the g_{et}. The mass of the unweighted distribution is much less concentrated about the center and much more concentrated in the tails. On an unweighted basis, about 25% of all manufacturing establishments experienced a growth rate in the interval [-.05,.05]. About 46% of all establishments experienced a growth rate in the interval [-.15,.15]. Births and deaths account for 14% of annual growth rate observations on manufacturing establishments. Evidently, establishment turnover and employment volatility is a sharply declining function of establishment size at annual frequencies. Dunne, Roberts, and Samuelson (1987,1989) show that this relationship also holds at low frequencies for U.S. manufacturing industries. Troske (1989) shows that this relationship holds at high frequencies in both manufacturing and nonmanufacturing industries. Evans (1987) and Hall (1987) find that employment volatility is a declining function of size in U.S. manufacturing firms, as opposed to establishments.

B. Cross-Industry and Time Variation in the Growth Rate Distribution

We now look at broad patterns of cross-industry and time variation in the size-weighted growth rate distributions. Figure 2 presents box plots that summarize key features of the establishment growth rate distribution for each two-digit manufacturing industry. Each box plot shows the mean and the 10th, 25th, 50th, 75th, and 90th percentiles of the size-weighted observations on the g_{et}. A solid horizontal line indicates the median, a dashed horizontal line indicates the mean, the height of the rectangle indicates the interquartile range, and the overall height of the box plot indicates the distance between the 90th and 10th percentiles. The box plots are ordered by the time-series variance of net industry employment growth rates.

Three interesting aspects of Figure 2 merit attention. First, the volatility of establishment growth rates varies greatly across industries. The most volatile industry, Lumber and Wood Products (SIC 24), shows an interquartile range of twenty-three percentage points. At the opposite end of the volatility spectrum, Petroleum Products (SIC 29) shows an interquartile range of only nine percentage points.

Second, scanning from left to right in Figure 2 reveals a strong positive relationship between the variability of the net industry employment growth rate and the heterogeneity of establishment-level employment changes within the industry. The cross-industry correlation between between the standard deviation of NET_{it} (from Table 3) and the interquartile range is .75 (marginal significance level of .0002 in a two-tailed test). Although difficult to discern in Figure 2, there is also a weak negative relationship between the secular industry growth rate and the heterogeneity of establishment-level employment changes within the industry. The cross-industry correlation between the mean and interquartile range is -.23.

Third, a striking aspect of Figure 2 is the tight clustering of industry median growth rates about zero. In contrast, mean growth rates differ significantly across industries. Rapidly contracting industries show means well below their medians and distributions that are skewed toward negative growth rates.[8]

Taken together, the relationships among the mean, median, and interquartile range indicate that the growth rate distributions for rapidly contracting industries do not simply represent mean-translated or median-translated versions of the growth rate distributions for slowly contracting industries. Instead, rapidly contracting industries typically exhibit growth rate densities that are more elongated, especially in the negative direction. Most of the cross-industry variation in secular contraction rates is accounted for by cross-industry differences in the number of establishments experiencing large contractions.

Figure 3 shows a time series of box plots. Each box plot shows the mean and selected percentiles of the size-weighted growth rate distribution for the indicated year. Consistent with our earlier finding of countercyclic variation in gross job reallocation, Figure 3 shows evidence of a negative time-series relationship between the mean establishment growth rate and the spread of the growth rate density.

Consider 1973, the year of most rapid employment expansion in our sample. This year exhibits the most concentrated growth rate density, and it is the only year for which the mean growth rate lies well above the median. The other years of positive growth–1976, 1977, and 1978–show roughly equal means and medians. 1977 and 1978, though not 1976, also show relatively concentrated growth rate densities. In contrast, during the years of sharp contraction–1975, 1982, 1983, 1985, and 1986–the mean establishment growth rate lies well below the median growth rate. Furthermore, these years show greater spread in their growth rate densities than 1973, 1977, 1978, or 1980. Thus, years of relatively

[8]Let Px define the xth percentile of the weighted empirical growth rate density, and consider the ratio (P90−P50)/(P50−P10). The cross-industry correlation between this ratio and the mean industry growth rate is .69 (.001).

rapid contraction in the manufacturing sector exhibit growth rate densities that are more elongated, especially in the negative direction.

C. Partitions of Job Creation and Destruction by Establishment Growth Rate

Figures 1A, 2, and 3 are constructed from size-weighted observations on establishment growth rates. To gauge the composition of gross employment changes, it is useful to further weight the establishment-level growth rate observations in proportion to their contribution to total job creation or destruction. The partitions of gross job creation and destruction described in section III.D accomplish this weighting.

Tables 4(a) and 4(b) partition gross job destruction and creation into seven intervals for each two-digit industry. For example, the rightmost column of Table 4(a) shows the fraction of gross job destruction accounted for by establishment deaths in the indicated industries. The other six columns show the contribution of less severe establishment-level contractions to gross job destruction. The intervals in Tables 4(a) and 4(b) are symmetric in terms of the g_{et} scale.

Tables 4 show considerable discreteness in establishment-level employment changes and illustrate the important role played by births and deaths in gross job creation and destruction activity. The fraction of gross job destruction accounted for by establishment deaths ranges from 15% in Transportation to 35% in Apparel. Severe contractions ($G_{et} \leq -.50$) account for 35% to 57% of gross job destruction, depending on industry. Excluding Tobacco, the fraction of gross job creation accounted for by establishment births ranges from 12% in Transportation to 28% in Apparel. Dramatic expansions ($G_{et} \geq 1.00$) account for 25% to 48% of gross job creation, depending on industry.

Small, incremental establishment-level employment changes also account for large fractions of gross job creation and destruction. Contractions of twenty percent or less (on the G_{et} scale) account for 15% to 37% of gross job destruction, depending on industry. Expansions of 25% or less (on the G_{et} scale) account for 21% to 52% of gross job creation, depending on industry.

Tables 5(a) and 5(b) present information on the time-series variation in the partitions of gross job destruction and creation. These tables show partitions for each year of the sample. Establishment deaths represent a significant fraction of gross job destruction during each year of the sample, but their relative importance varies from year to year. In 1977, deaths accounted for 36% of gross job destruction, while severe contractions accounted for 52% of gross job destruction. In 1980, deaths accounted for 14% of gross job destruction, while severe contractions accounted for 34% of gross job destruction.

Establishment births represent a more modest, but still significant, fraction of gross job creation. The contribution of establishment births to gross job creation ranges from

8% in 1981 to 33% in 1975. Dramatic expansions account for 23% to 47% of gross job creation, depending on year.

D. Summary

To conclude this section, we summarize the three most important descriptive characterizations of the establishment growth rate densities and the partitions of gross job creation and destruction. First, establishment births and deaths account for only 2.4% of the size-weighted observations on annual growth rates, but they account for 14% of the observations on an unweighted basis. More generally, establishment growth rate volatility is a sharply declining function of establishment size in annual data.

Second, in both a time-series and cross-industry sense, lower mean establishment growth rates are associated with establishment growth rate densities that exhibit greater spread and greater skewness. With respect to skewness, the mean establishment growth rate typically lies well below the median growth rate in industries experiencing sharp secular contractions and in the manufacturing sector during years of rapid contraction.

Third, establishment-level employment changes show considerable discreteness. Establishment deaths and severe contractions account for more than a third of gross job destruction in every two-digit industry, on average, and for the manufacturing sector in every year of the sample. Establishment births and dramatic expansions also account for a large fraction of gross job creation. Our finding of considerable discreteness in establishment-level employment changes is consistent with evidence in Hamermesh (1989) on monthly employment changes in a small sample of manufacturing plants.

VI. Accounting for Fluctuations in Gross Job Reallocation: Methodology

A. An Outline of the Methodology

Our methodology to account for time variation in gross job creation, destruction and reallocation proceeds in five steps. We first specify simple models of establishment-level growth rates that are linear in time-varying aggregate, sectoral and idiosyncratic effects. Second, and in accordance with the establishment-level growth rate model, we remove the contribution of aggregate and/or sectoral effects to time variation in the shape and location of the emprical density over establishment growth rates. Third, we recompute the time series on gross job creation, destruction and reallocation using the adjusted time series on the establishment growth rate density. Fourth, we apply simple variance decompositions to quantify the contribution of aggregate, sectoral and idiosyncratic effects to the time variation in gross job creation, destruction and reallocation. Finally, we examine the time-series behavior of the part of gross job reallocation associated with the idiosyncratic

component. We now describe our methodology and establishment-level growth rate models in greater detail.

B. The Establishment-Level Growth Rate Model

Consider the linear model for establishment-level growth rates

$$g_{et} = \tilde{g}_{et}^{ST} + g_{st} + g_t, \tag{1}$$

where:

$g_t = \Delta EMP_t / X_t$, an aggregate-year effect;

$g_{st} = \Delta EMP_{st} / X_{st} - g_t$, a sector-year effect (deviated about the aggregate-year effect);

$\tilde{g}_{et}^{ST}$ = the year-t growth rate for establishment e deviated about the aggregate-year and sector-year effects.

We also define:

$\tilde{g}_{et}^{T} = g_{et} - g_t$, the year-$t$ growth rate for establishment e deviated about the aggregate-year effect; and

$\tilde{g}_{et}^{S} = g_{et} - g_{st}$, the year-$t$ growth rate for establishment e deviated about the sector-year effect.

According to equation (1), each establishment's growth rate at t is the sum of an aggregate-year effect, a sector-year effect and a time-varying idiosyncratic effect. The aggregate-year effect is simply the realized (size-weighted) mean value of the establishment growth rates. The sector-year effects are the realized sectoral mean values of the establishment growth rates, deviated about the aggregate mean. Time variation in the realized aggregate and sectoral means induce time variation in the location and shape of the density over the (size-weighted) g_{et}, thereby generating time variation in gross job creation, destruction and reallocation. The cross-sectional variance and higher moments of the idiosyncratic component, $\tilde{g}_{et}^{ST}$, also influence the shape of the growth rate density, thereby generating time variation in gross job creation, destruction and reallocation.

In terms of equation (1), a major objective of our empirical methodology is to apportion the time variation in gross job creation, destruction, and reallocation among three factors: (a) time variation in the realized values of g_t; (b) time variation in the realized values of the g_{st}, $s = 1, \ldots S$; and (c) time variation in the realized cross-sectional variance and higher moments of the distribution over the $\tilde{g}_{et}^{ST}$.

C. Accounting for Fluctuations: Aggregate Versus Idiosyncratic Effects

Several alternative views about the nature of aggregate fluctuations can be couched in terms of equations like (1). Prevailing views of the business cycle stress the role of aggregate disturbances as driving forces. The simplest version of this view implies that all

time variation in gross job creation, destruction and reallocation is driven by time variation in the aggregate-year effects. This view encompasses a time-invariant, but possibly large, cross-sectional variance of the idiosyncratic component of the g_{et}. We represent this pure aggregate shifts story by the hypothesis that the distribution over the $\tilde{g}_{et}^T = g_{et} - g_t$ is time invariant. Our empirical analysis investigates the aspects of this hypothesis that pertain to the time-series behavior of gross job creation, destruction and reallocation.

From the distribution over the $\tilde{g}_{et}^T$, we compute the gross job creation, destruction and reallocation rates adjusted for the aggregate-year effects:

$$\widetilde{POS}_t^T = \frac{1}{2}\sum_e \frac{x_{et}}{X_t}(|\tilde{g}_{et}^T| + \tilde{g}_{et}^T), \tag{2}$$

$$\widetilde{NEG}_t^T = \frac{1}{2}\sum_e \frac{x_{et}}{X_t}(|\tilde{g}_{et}^T| - \tilde{g}_{et}^T), \qquad \text{and,} \tag{3}$$

$$\widetilde{SUM}_t^T = \sum_e \frac{x_{et}}{X_t}|\tilde{g}_{et}^T|. \tag{4}$$

The time-series movements in these adjusted measures reflect the contributions of the realized values of the sectoral mean effects and the higher moments of the idiosyncratic effects. Note that since $\sum_e (x_{et}/X_t)\tilde{g}_{et}^T$ equals zero by definition, $2\widetilde{POS}_t^T \equiv 2\widetilde{NEG}_t^T \equiv \widetilde{SUM}_t^T$. In other words, after removing the contribution of the aggregate-year effects, the gross job creation, destruction and reallocation measures have identical time-series properties.

Now consider the identity

$$SUM_t = \widetilde{SUM}_t^T + (SUM_t - \widetilde{SUM}_t^T), \tag{5}$$

which implies the variance decomposition for gross job reallocation,

$$\text{Var}(SUM_t) = \text{Var}(\widetilde{SUM}_t^T) + \text{Var}(SUM_t - \widetilde{SUM}_t^T) + 2\text{Cov}(\widetilde{SUM}_t^T, SUM_t - \widetilde{SUM}_t^T). \tag{6}$$

If the distribution over the $\tilde{g}_{et}^T$ is time-invariant, then the ratio of $\text{Var}(\widetilde{SUM}_t^T)$ to $\text{Var}(SUM_t)$ equals zero. Conversely, a large value for this ratio indicates that time variation in the cross-sectional variance (and higher moments) of $\tilde{g}_{et}^T$ accounts for much of the time variation in gross job reallocation. A large value for the ratio of $\text{Var}(SUM_t - \widetilde{SUM}_t^T)$

to Var(SUM_t) indicates that time variation in g_t accounts for much of the time variation in gross job reallocation. We interpret the covariance term as reflecting the part of time variation in gross job reallocation that cannot be unambiguously assigned to either the aggregate or sectoral and idiosyncratic effects.

We similarly decompose the variance of gross job creation and destruction rates along the lines of equations (5) and (6). Variance ratios provide information on the relative contribution of aggregate and idiosyncratic effects to time variation in job creation and destruction. The covariance terms indicate whether the idiosyncratic effects reinforce or counteract the aggregate effects in terms of their contributions to gross job creation and destruction.

D. Accounting for Fluctuations: More Sophisticated Characterizations of Traditional Views

A less simplistic characterization of prevailing views about the business cycle would recognize perennial differences in the timing and magnitude of sectoral responses to aggregate disturbances. These cross-sectoral differences in the responses to aggregate disturbances are an important element of traditional views about the business cycle. See Abraham and Katz (1986) on this point.

To capture this aspect of traditional views, we could proceed by explicitly modelling the differential cross-sectoral responses to aggregate disturbances. We would then obtain an analog to the distribution over the $\tilde{g}_{et}^{T}$ and investigate implications of the hypothesis that this new distribution is time invariant. While this approach has its virtues, it would too closely tie our empirical analysis to one or more particular representations of the link between aggregate disturbances and sectoral employment responses. Our alternative approach entails two main elements: first, we allow for completely unrestricted sectoral responses to aggregate disturbances; second, we consider several sectoral classification schemes.

Consider the hypothesis of a time-invariant distribution over the $\tilde{g}_{et}^{ST}$. Note that the sector-year effects, g_{st}, capture *any* systematic cross-sectoral differences in the response to aggregate disturbances. (Of course, they capture any non-systematic differences as well.) Neither linearity, magnitude, nor timing restrictions are placed on the sectoral responses to aggregate disturbances under this interpretation of the g_{st}. The only restrictions placed on the sectoral responses are those inherent in the sectoral classification scheme itself.

Now consider the adjusted gross job reallocation measure, $\widetilde{SUM}_t^{ST}$, constructed from the distribution over the $\tilde{g}_{et}^{ST}$:

$$\widetilde{SUM}_t^{ST} = \sum_e \frac{x_{et}}{X_t} |\tilde{g}_{et}^{ST}| \qquad (4').$$

As a measure of the intensity of shifts in the pattern of employment opportunities, this adjusted gross job reallocation measure is immune to the criticisms that Abraham and Katz (1986) directed towards Lilien's (1982a) dispersion measure, because it accomodates arbitrary sectoral responses to aggregate disturbances.[9] *That is, conditional on the sectoral classification scheme,* $\widetilde{SUM}_t^{ST}$ *is a true measure of the idiosyncratic contribution to gross job reallocation. Accordingly, its comovement with net sectoral and aggregate growth provides direct evidence on the hypothesis that the idiosyncratic contribution to gross job reallocation is countercyclic.*

To characterize the cyclic behavior of various adjusted gross job reallocation measures, we investigate their comovement with net sectoral and manufacturing employment growth rates using both regression and correlation analyses.

VII. Accounting for Fluctuations in Gross Job Reallocation: Results

A. Variance Decompositions

Table 6 reports variance decompositions based on several alternative sectoral classification schemes. The time-series variance of the raw measures appears in panel A. The next seven panels report variance ratios corresponding to various empirical densities, as adjusted for aggregate-year effects and/or sector-year effects. The superscripts on X_t indicate the particular empirical density used to construct the adjusted job reallocation and related measures. For example, the "T" superscript in panel B indicates that the adjusted measures are constructed from the density over the $\tilde{g}_{et}^T$, as in (2)-(4). In panel C, the "$I2$" superscript indicates that the density is adjusted for two-digit industry-year effects only. The remaining panels correspond to densities adjusted for both aggregate-year effects and sector-year effects. Panel D corresponds to the two-digit industry scheme. Panel E defines sectors in terms of the roughly 450 four-digit industries. Panel F defines sectors in terms of the nine Census geographic regions. Panel G defines sectors in terms of three establishment size classes.[10] Panel H defines sectors in terms of two-digit industry/regions. Finally, in panel I we delete observations on establishments during the year of birth and/or death and define sectors in terms of two-digit industries.

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[9] Of course, if we allow for arbitrary sectoral responses to aggregate disturbances, we cannot separately identify the contribution of sectoral disturbances, as distinct from aggregate disturbances, to fluctuations in gross job creation, destruction and reallocation.

[10] We define the size of establishment e as the simple mean over all observations on x_{et}. We then designate the establishment as small, medium or large depending on whether its measure of size falls in the interval (0,249], (249,999] or (999,∞).

The column headed SUM in Table 6 reports the variance ratios for gross job reallocation. According to Panel B, aggregate-year effects alone unambiguously account for only three percent of the time variation in gross job reallocation, as indicated by the ratio of $\text{Var}(SUM_t - \widetilde{SUM}_t^T)$ to $\text{Var}(SUM_t)$. Given that aggregate-year effects represent simple mean translations of the empirical density, and given the definition of gross job reallocation, this finding is unsurprising. More surprising, panel C indicates that two-digit industry-year effects alone account for a mere .1% of the time variation in gross job reallocation. Panel D shows that, even in combination, aggregate-year effects and two-digit industry-year effects unambiguously account for less than five percent of time variation in gross job reallocation. Turning to panel E, which involves 450 distinct industry effects per year, the aggregate-year effects and sector-year effects unambiguously contribute only 10.5% of the variability of gross job reallocation. The results reported in panels F-I are even less supportive of the hypothesis that aggregate-year effects and sector-year effects account for significant time variation in gross job reallocation.

In contrast to the anemic role of aggregate-year effects and sector-year effects, idiosyncratic effects unambiguously account for 80% or more of the variability of gross job reallocation, regardless of the sectoral classification scheme and regardless of whether we restrict the sample to continuers. Returning to panel E, even when we allow for one aggregate and 450 distinct industry effects per year, idiosyncratic effects account for 80% of the time variation in gross job reallocation.

We interpret these variance ratio results as a decisive rejection of the hypothesis that the normal pattern of sectoral responses to aggregate fluctuations can account for significant movements in gross job reallocation. Instead, the time variation in gross job reallocation results overwhelmingly from time variation in the contribution of idiosyncratic effects. The results are especially striking in that our definition of idiosyncratic effects imposes neither linearity, magnitude nor timing restrictions on the mean sectoral responses to aggregate disturbances.

Turning to the columns headed POS and NEG in Table 6, aggregate-year effects play the major role in accounting for fluctuations in job creation and destruction rates. The variance of the idiosyncratic component of job creation amounts to only 12-16% of the variance of the raw job creation measure. The variance of the idiosyncratic component of the job destruction measure amounts to only 6-8% of the variance of the raw job destruction measure.

However, these variance ratios alone provide an incomplete picture of the cyclic behavior of gross job creation and destruction. Indeed, the second and third rows in panels

B-I tell an interesting story about the interaction between aggregate and idiosyncratic effects in accounting for fluctuations in job creation and destruction. Aggegate and sectoral effects alone account for only about 65% of fluctuations in job destruction rates. The covariance of the aggregate and sectoral effects with the idiosyncratic effects accounts for most of the remaining fluctuations in job destruction. The positive sign and nontrivial magnitude of the covariance term indicate that idiosyncratic effects strongly reinforce the countercyclic fluctuations in gross job creation associated with mean aggregate effects.

The covariance terms plays an even more important role in the variance decompositions for job creation. Here the idiosyncratic effects strongly counteract the influence of aggregate effects on job creation. In the absence of idiosyncratic effects, Table 6 indicates that the variance of job creation would be roughly 40% larger than the observed variance.

In sum, Table 6 supports the following conclusions. First, idiosyncratic effects account for the overwhelming bulk of the significant time variation in gross job reallocation. Second, idiosyncratic effects strongly reinforce the contribution of mean aggregate effects to the countercyclic variation of gross job destruction. Third, idiosyncratic effects strongly counteract the contribution of mean aggregate effects to the procyclic fluctuations in gross job creation.

B. Time-Series Correlations between Net Job Growth and Gross Job Reallocation

Table 7 summarizes the pattern of contemporaneous correlations between own-sector net job growth and various job reallocation measures. Although not shown here, results are similar for correlations between sectoral job reallocation measures and manufacturing net job growth.

The basic pattern in Table 7 is clear: both raw and adjusted sectoral gross job reallocation measures fluctuate countercyclically. For example, defining sectors as two-digit industries, the size-weighted mean time-series correlation between net industry job growth and own-industry job reallocation equals -.51. Adjusting the empirical growth rate density for aggregate and sectoral effects yields a mean correlation of -.55. Furthermore, all twenty two-digit industries exhibit a negative time-series correlation between net job growth and the raw and adjusted job reallocation measures.

A similar pattern prevails when the sample is restricted to continuing establishments and for the other sectoral classification schemes. Mean correlations fall to about -.4 when sectors are defined in terms of the roughly 180 two-digit industry/regions and to about -.35 when defined in terms of the roughly 450 four-digit industries. All nine geographic regions, the three establishment size classes, 80% of the four-digit industries, and 85% of the two-digit industry/regions show a negative correlation between net job growth and adjusted gross job reallocation.

Although not shown in Table 7, the correlations for individual size-class sectors reveal an interesting relationship: countercyclic variation in gross job reallocation is much more pronounced for large establishments than for small establishments. The time-series correlation (marginal significance level) between net job growth and gross job reallocation (adjusted for aggregate-year and size-year effects) is -.15 (.66) for the small establishment sector, -.60 (.05) for the mid-sized establishment sector, and -.84 (.001) for the large establishment sector. However, average rates of gross job reallocation are greater for small establishments, as indicated by the comparison between Figures 1(A) and 1(B). Thus, while small establishments account for a disproportionate fraction of the underlying gross job reallocation rate, large establishments account for a disproportionate fraction of the countercyclic variation in job reallocation.[11]

C. Job Reallocation Regressions

To further investigate the pattern of comovement between gross job reallocation and net job growth, we examine OLS regressions of $\widetilde{SUM}_{st}^{ST}$ on sectoral and aggregate growth rates. The regressions include sector fixed effects to control for sytematic cross-sectoral differences in the rate of job reallocation. Results appear in Table 8. For reasons described in the introduction (section I.E.), we do not impute a structural or causal interpretation to these regressions. Instead, we use them to gauge the magnitude and significance of the time-series covariance between gross job reallocation and net job growth, while controlling for permanent cross-sectoral differences in gross job reallocation.

Table 8 shows a statistically signficant inverse relationship between the idiosyncratic component of gross job reallocation and net job growth at both the aggregate and sectoral levels.[12] This relationship holds for all sectoral classification schemes except size, where we find a statistically significant inverse relationship only for aggregate net growth.

Table 8 also shows some evidence of important differences between the coefficient estimates on own sector and aggregate net job growth. Under the 4-digit, 2-digit/region, and region classification schemes, declines in own-sector net growth are associated with significantly (in both economic and statistical terms) greater gross job reallocation increases than declines in manufacturing net growth. The other classification schemes show

[11] The mean annual rates (standard deviation) of gross job reallocation over time, by size-class, are as follows: small, .27 (.02); medium, .18 (.02); and large, .14 (.02).

[12] All standard errors and test statistics in Table 8 are based on White's heteroscedasticity-consistent covariance matrix estimator. Results based on the standard OLS covariance matrix estimator, and results based on WLS (weight proportional to the square root of the number of establishments in the sector) are in all essential respects identical.

no statistically significant difference between the estimated coefficients on own-sector and manufacturing net growth.

Using the results in Table 8, we can summarize the magnitude of the covariance between net job growth and gross job reallocation as follows. Evaluated at sample means, a one standard deviation decrease in g_t is associated with an increase in $\widetilde{SUM}_t^{I2T}$ from 20.2% to 21.3%, an increase in $\widetilde{SUM}_t^{I4T}$ from 20.9% to 21.7%, an increase in $\widetilde{SUM}_t^{ZT}$ from 19.5% to 20.7%, an increase in $\widetilde{SUM}_t^{RT}$ from 21.0% to 21.9%, an increase in $\widetilde{SUM}_t^{I2RT}$ from 21.6% to 22.4%, and an increase in $\widetilde{SUM}_t^{I2T}$ (continuing establishments only) from 15.6% to 16.7%. Likewise, a one standard deviation decrease in g_{st} is associated with an increase in $\widetilde{SUM}_t^{I2T}$ from 20.2% to 21.1%, an increase in $\widetilde{SUM}_t^{I4T}$ from 20.9% to 23.3%, an increase in $\widetilde{SUM}_t^{RT}$ from 21.0% to 22.1%, an increase in $\widetilde{SUM}_t^{I2RT}$ from 21.6% to 23.8%, and an increase in $\widetilde{SUM}_t^{I2T}$ (continuing establishments only) from 15.6% to 16.3%. Comparing these effects with the standard deviation of $\widetilde{SUM}_t^{ST}$ for each sector reveals that much of the time variation in the idiosyncratic component of gross job reallocation is accounted for by time variation in aggregate and own-sector net growth.

It is useful to place these findings alongside our earlier variance decomposition results. The variance decomposition results show that the great bulk of time variation in gross job reallocation is explained by idiosyncratic effects on the shape of the empirical growth rate density. The correlation and regression results show that the contributon of idiosyncratic effects to time variation in the shape of the density leads to large and systematic countercyclic variation in gross job reallocation. Taken together, these findings provide strong evidence that net aggregate and sectoral employment fluctuations are intimately related to fluctuations in the intensity of shifts in employment opportunities across establishments.

VIII. Concluding Remarks

Establishment-level employment changes exhibit tremendous heterogeneity, even within narrowly defined sectors of the economy. In this respect, our empirical results parallel Roll's (1988) results on the tremendous, and largely unexplained, heterogeneity of firm-level stock returns. In labor economics, the tremendous heterogeneity of earnings and labor supply behavior has been evident since the advent of empirical studies using micro data. Similarly, studies of industrial organization reveal profound differences across firms and establishments in even such basic characteristics as size. In short, heterogeneity is a pronounced feature of economic phenomena.

In the literature on aggregate economic fluctuations, researchers have typically abstracted from much or even all of this heterogeneity. A recurrent theme in modern business cycle research is the idea that the essential features of aggregate fluctuations can be represented and explained within stochastic growth models inhabited by representative consumer-workers and firms. This theme, clearly annuciated in Lucas (1977), is most pronounced in the burgeoning literature on real business cycle analysis à la Kydland and Prescott (1982).[13]

Abstraction from inessential considerations is one hallmark of successful theories, and abstraction from firm and worker heterogeneity has often proved to be a useful simplifying device in macroeconomics. In this respect, our work in no way denies the insights that emerge from business cycle analyses that rely on the representative agent paradigm or other aggregation devices. But our finding of significant countercyclic variation in gross job reallocation suggests that analyses based on representative agent models are likely to yield seriously incomplete interpretations and explanations for aggregate economic fluctuations. While it is certainly conceivable that the significant countercyclic variation in gross job reallocation that we document plays no important role in aggregate fluctuations, we find such a view difficult to square with modern views of labor market dynamics and their emphasis on the nontrivial costs associated with job loss, worker reallocation, and specific capital formation.

[13]This theme is prevalent, but by no means universal, in modern business cycle research. The cyclic behavior of real wages, for example, is one area where the implications of heterogeneity have recently received serious attention. See Bils (1985) and Keane, Moffitt and Runkle (1988).

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TABLE 1

Net and Gross Rates by Industry

Size-Weighted Averages:[1]

Industry:		$\overline{POS}_{it}$	$\overline{NEG}_{it}$	$\overline{NET}_{it}$	$\overline{SUM}_{it}$	$\overline{MAX}_{it}$
Food	(20)	0.089	0.104	-0.015	0.193	0.108
Tobacco	(21)	0.058	0.082	-0.024	0.140	0.090
Textile	(22)	0.074	0.110	-0.036	0.185	0.124
Apparel	(23)	0.116	0.156	-0.040	0.272	0.168
Lumber	(24)	0.129	0.160	-0.031	0.288	0.188
Furniture	(25)	0.101	0.121	-0.019	0.222	0.143
Paper	(26)	0.063	0.078	-0.015	0.141	0.089
Printing	(27)	0.091	0.087	-0.004	0.178	0.099
Chemicals	(28)	0.068	0.080	-0.013	0.148	0.089
Petroleum	(29)	0.066	0.091	-0.025	0.157	0.100
Rubber	(30)	0.107	0.118	-0.011	0.225	0.143
Leather	(31)	0.091	0.144	-0.053	0.235	0.152
Stone, Clay and Glass	(32)	0.093	0.123	-0.031	0.216	0.136
Primary Metals	(33)	0.059	0.114	-0.054	0.173	0.126
Fabricated Metals	(34)	0.095	0.120	-0.025	0.215	0.137
Nonelectric Machinery	(35)	0.096	0.121	-0.025	0.217	0.141
Electric Machinery	(36)	0.097	0.109	-0.011	0.206	0.130
Transportation	(37)	0.094	0.099	-0.006	0.193	0.123
Instruments	(38)	0.093	0.093	-0.002	0.186	0.112
Miscellaneous	(39)	0.108	0.145	-0.037	0.253	0.156
Total Manufacturing		0.092	0.113	-0.021	0.205	0.129
Size-weighted Cross-Industry Standard Deviation		0.011	0.015	0.011	0.024	0.016

Pearson correlations:[2] $\rho(\overline{POS}_{it}, \overline{NEG}_{it}) = 0.764$ (0.0001) $\quad \rho(\overline{NET}_{it}, \overline{SUM}_{it}) = -0.347$ (0.135)

Rank correlations: $\rho(\overline{POS}_{it}, \overline{NEG}_{it}) = 0.719$ (0.0004) $\quad \rho(\overline{NET}_{it}, \overline{SUM}_{it}) = -0.346$ (0.135)

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[1]Size-weighted average based on annual values with t = 1973-1986 (excluding 1974, 1979, 1984).

[2]Marginal significance levels in parentheses.

TABLE 2

Net and Gross Rates by Year for Total Manufacturing

	POS_t	NEG_t	NET_t	SUM_t	MAX_t
1973	0.132	0.061	0.071	0.194	0.133
1975	0.067	0.166	-0.100	0.233	0.166
1976	0.113	0.096	0.017	0.209	0.122
1977	0.112	0.096	0.018	0.206	0.117
1978	0.116	0.075	0.041	0.191	0.117
1980	0.080	0.093	-0.012	0.173	0.102
1981	0.070	0.118	-0.049	0.188	0.119
1982	0.064	0.152	-0.087	0.216	0.152
1983	0.086	0.142	-0.056	0.227	0.143
1985	0.084	0.117	-0.033	0.201	0.121
1986	0.088	0.132	-0.044	0.220	0.133

TABLE 3

Time Series Variation in Net and Gross Rates by Industry

Time Series Standard Deviation of:

Industry:		POS_{it}	NEG_{it}	NET_{it}	SUM_{it}
Food	(20)	0.014	0.016	0.024	0.018
Tobacco	(21)	0.020	0.028	0.046	0.019
Textile	(22)	0.031	0.046	0.071	0.031
Apparel	(23)	0.036	0.041	0.072	0.028
Lumber	(24)	0.048	0.059	0.102	0.035
Furniture	(25)	0.040	0.050	0.083	0.036
Paper	(26)	0.019	0.031	0.048	0.017
Printing	(27)	0.017	0.018	0.027	0.021
Chemicals	(28)	0.014	0.020	0.032	0.014
Petroleum	(29)	0.023	0.033	0.046	0.034
Rubber	(30)	0.036	0.048	0.082	0.023
Leather	(31)	0.027	0.052	0.072	0.042
Stone, Clay and Glass	(32)	0.027	0.042	0.063	0.031
Primary Metals	(33)	0.025	0.063	0.082	0.049
Fabricated Metals	(34)	0.029	0.041	0.068	0.021
Nonelectric Machinery	(35)	0.031	0.051	0.077	0.032
Electric Machinery	(36)	0.028	0.047	0.073	0.027
Transportation	(37)	0.023	0.040	0.060	0.027
Instruments	(38)	0.020	0.032	0.047	0.026
Miscellaneous	(39)	0.031	0.038	0.065	0.023
Total Manufacturing		0.023	0.031	0.052	0.016
Size-weighted Industry Average		0.027	0.040	0.062	0.027

TABLE 4(a)

Partition of Gross Job Destruction by Industry

Size-Weighted Average of $NEG_{it}[X,Y)$:[1,2]
$[X,Y) =$

g_{et} scale:		[0,.095)	[.095,.222)	[.222,.40)	[.40,.666)	[.666,1.439)	[1.439,2)	[2,2]
G_{et} scale:		[0,.091)	[.091,.20)	[.20,.333)	[.333,.50)	[.50,.833)	[.833,1.00)	[1.00,1.00]
Industry:								
Food	(20)	0.09	0.16	0.13	0.12	0.13	0.05	0.31
Tobacco	(21)	0.14	0.23	0.15	0.13	0.11	0.09	0.20
Textile	(22)	0.10	0.18	0.17	0.12	0.12	0.05	0.26
Apparel	(23)	0.06	0.12	0.14	0.13	0.14	0.06	0.35
Lumber	(24)	0.04	0.11	0.14	0.14	0.18	0.07	0.32
Furniture	(25)	0.07	0.17	0.18	0.14	0.12	0.03	0.30
Paper	(26)	0.13	0.21	0.17	0.11	0.08	0.04	0.26
Printing	(27)	0.10	0.16	0.14	0.13	0.12	0.03	0.32
Chemicals	(28)	0.13	0.21	0.14	0.13	0.12	0.04	0.23
Petroleum	(29)	0.13	0.13	0.12	0.13	0.22	0.06	0.21
Rubber	(30)	0.07	0.17	0.18	0.16	0.14	0.03	0.25
Leather	(31)	0.07	0.16	0.17	0.13	0.13	0.04	0.32
Stone, Clay and Glass	(32)	0.07	0.17	0.17	0.14	0.15	0.04	0.26
Primary Metals	(33)	0.09	0.20	0.20	0.16	0.13	0.05	0.16
Fabricated Metals	(34)	0.07	0.18	0.20	0.17	0.14	0.03	0.21
Nonelectric Machinery	(35)	0.06	0.19	0.20	0.17	0.14	0.02	0.21
Electric Machinery	(36)	0.08	0.19	0.20	0.17	0.13	0.03	0.20
Transportation	(37)	0.08	0.20	0.22	0.16	0.16	0.04	0.15
Instruments	(38)	0.10	0.20	0.17	0.13	0.10	0.02	0.28
Miscellaneous	(39)	0.06	0.15	0.16	0.15	0.13	0.05	0.31
Total Manufacturing		0.08	0.17	0.18	0.15	0.14	0.04	0.25

[1] t = 1973-1986 (excluding 1974, 1979, and 1984).

[2] $NEG_{it}[X,Y)$ is the fraction of NEG_{it} accounted for by establishments with employment decreases in the interval $[X,Y)$.

TABLE 4(b)

Partition of Gross Job Creation by Industry

Size-Weighted Average of $POS_{it}[X,Y)$:[1,2]
$[X,Y)$ =

g_{et} scale:		[0,.095)	[.095,.222)	[.222,.40)	[.40,.666)	[.666,1.439)	[1.439,2)	[2,2]
G_{et} scale:		[0,.10)	[.10,.25)	[.25,.50)	[.50,1.00)	[1.00,5.00)	[5.00,∞)	[∞,∞]
Industry:								
Food	(20)	0.11	0.21	0.18	0.13	0.12	0.05	0.20
Tobacco	(21)	0.16	0.38	0.10	0.12	0.17	0.02	0.06
Textile	(22)	0.13	0.23	0.17	0.13	0.10	0.03	0.19
Apparel	(23)	0.07	0.16	0.16	0.13	0.13	0.07	0.28
Lumber	(24)	0.06	0.15	0.17	0.15	0.16	0.06	0.25
Furniture	(25)	0.09	0.21	0.19	0.14	0.11	0.03	0.22
Paper	(26)	0.19	0.25	0.16	0.10	0.08	0.02	0.20
Printing	(27)	0.12	0.20	0.16	0.12	0.11	0.03	0.26
Chemicals	(28)	0.17	0.24	0.16	0.11	0.11	0.03	0.19
Petroleum	(29)	0.15	0.18	0.14	0.10	0.22	0.04	0.16
Rubber	(30)	0.08	0.20	0.21	0.15	0.12	0.04	0.20
Leather	(31)	0.09	0.20	0.23	0.17	0.10	0.02	0.19
Stone, Clay and Glass	(32)	0.09	0.20	0.18	0.13	0.13	0.03	0.23
Primary Metals	(33)	0.16	0.25	0.20	0.11	0.09	0.03	0.16
Fabricated Metals	(34)	0.09	0.22	0.21	0.16	0.11	0.02	0.19
Nonelectric Machinery	(35)	0.09	0.23	0.20	0.15	0.10	0.02	0.20
Electric Machinery	(36)	0.10	0.24	0.22	0.14	0.10	0.02	0.18
Transportation	(37)	0.12	0.26	0.23	0.14	0.11	0.03	0.12
Instruments	(38)	0.11	0.24	0.20	0.12	0.08	0.04	0.22
Miscellaneous	(39)	0.07	0.18	0.20	0.16	0.13	0.03	0.23
Total Manufacturing		0.11	0.22	0.19	0.14	0.11	0.03	0.20

[1] t = 1973-1986 (excluding 1974, 1979, and 1984).
[2] $POS_{it}[X,Y)$ is the fraction of POS_{it} accounted for by establishments with employment increases in the interval $[X,Y)$.

TABLE 5(a)

Partition of Gross Job Destruction by Year

Size-Weighted Average of $NEG_t[X,Y)$:[1,2]

g_{et} scale:	[0,.095)	[.095,.222)	[.222,.40)	[.40,.666)	[.666,1.439)	[1.439,2)	[2,2]
G_{et} scale:	[0,.091)	[.091,.20)	[.20,.333)	[.333,.50)	[.50,.833)	[.833,1.00)	[1.00,1.00]
Year:							
1973	0.11	0.19	0.15	0.10	0.10	0.04	0.31
1975	0.06	0.17	0.21	0.20	0.15	0.04	0.17
1976	0.08	0.19	0.19	0.13	0.10	0.03	0.28
1977	0.08	0.15	0.13	0.11	0.12	0.04	0.36
1978	0.10	0.17	0.14	0.11	0.13	0.04	0.31
1980	0.10	0.20	0.20	0.16	0.15	0.05	0.14
1981	0.09	0.19	0.18	0.16	0.14	0.04	0.19
1982	0.07	0.18	0.19	0.15	0.13	0.04	0.25
1983	0.07	0.17	0.18	0.17	0.16	0.05	0.20
1985	0.08	0.14	0.15	0.15	0.15	0.04	0.29
1986	0.07	0.16	0.16	0.13	0.13	0.03	0.32

[1]Size Weighted Average for Total Manufacturing.
[2]$NEG_t[X,Y)$ is the fraction of NEG_t accounted for by establishments with employment decreases in the interval [X,Y).

TABLE 5(b)

Partition of Gross Job Creation by Year

Size-Weighted Average of $POS_t[X,Y)$:[1,2]

g_{et} scale:	[0,.095)	[.095,.222)	[.222,.40)	[.40,.666)	[.666,1.439)	[1.439,2)	[2,2]
G_{et} scale:	[0,.10)	[.10,.25)	[.25,.50)	[.50,1.00)	[1.00,5.00)	[5.00,∞)	[∞,∞]
Year:							
1973	0.09	0.23	0.20	0.13	0.09	0.03	0.22
1975	0.10	0.18	0.16	0.11	0.14	0.04	0.29
1976	0.08	0.19	0.24	0.18	0.13	0.03	0.14
1977	0.10	0.23	0.19	0.14	0.12	0.03	0.18
1978	0.11	0.22	0.19	0.13	0.10	0.03	0.21
1980	0.13	0.27	0.22	0.16	0.11	0.04	0.08
1981	0.13	0.24	0.18	0.15	0.10	0.04	0.17
1982	0.12	0.22	0.18	0.14	0.15	0.03	0.16
1983	0.09	0.19	0.15	0.13	0.11	0.06	0.27
1985	0.12	0.22	0.21	0.12	0.11	0.02	0.19
1986	0.11	0.20	0.16	0.10	0.08	0.02	0.33

[1]Size Weighted Average for Total Manufacturing.
[2]$POS_t[X,Y)$ is the fraction of POS_t accounted for by establishments with employment increases in the interval [X,Y).

TABLE 6

Variance Ratios[1,2]

X =

PANEL		SUM	POS	NEG
A	$V(X_t)$	0.00033	0.00052	0.00107
	$V(X_t)$ (continuing establishments only)	0.00019	0.00039	0.00088
B	$V(\bar{X}^T_t)/V(X_t)$	1.026	0.16	0.079
	$V(X_t-\bar{X}^T_t)/V(X_t)$	0.03	1.44	0.63
	$2COV(\bar{X}^T_t, X_t-\bar{X}^T_t)/V(X_t)$	-0.056	-0.60	0.287
C	$V(\bar{X}^{I2}_t)/V(X_t)$	0.95	0.995	1.0017
	$V(X_t-\bar{X}^{I2}_t)/V(X_t)$	0.001	0.0002	0.0001
	$2COV(\bar{X}^{I2}_t, X_t-\bar{X}^{I2}_t)/V(X_t)$	0.013	0.0048	-0.0018
D	$V(\bar{X}^{I2T}_t)/V(X_t)$	0.876	0.136	0.068
	$V(X_t-\bar{X}^{I2T}_t)/V(X_t)$	0.044	1.395	0.658
	$2COV(\bar{X}^{I2T}_t, X_t-\bar{X}^{I2T})/V(X_t)$	0.079	-0.531	0.274
E	$V(\bar{X}^{I4T}_t)/V(X_t)$	0.797	0.124	0.062
	$V(X_t-\bar{X}^{I4T}_t)/V(X_t)$	0.105	1.318	0.705
	$2COV(\bar{X}^{I4T}_t, X_t-\bar{X}^{I4T}_t)/V(X_t)$	0.098	-0.442	0.233
F	$V(\bar{X}^{RT}_t)/V(X_t)$	1.034	0.16	0.080
	$V(X_t-\bar{X}^{RT}_t)/V(X_t)$	0.035	1.44	0.637
	$2COV(\bar{X}^{RT}_t, X_t-\bar{X}^{RT}_t)/V(X_t)$	-0.069	-0.60	0.283

Table 6 (cont.)

PANEL		SUM	POS	NEG
	$V(\bar{X}^{Z}_{t})/V(X_{t})$	1.032	0.160	0.080
G	$V(X_{t}-\bar{X}^{Z}_{t})/V(X_{t})$	0.035	1.448	0.632
	$2COV(\bar{X}^{Z}_{t},X_{t}-\bar{X}^{Z}_{t})/V(X_{t})$	-0.067	-0.607	0.288
	$V(\bar{X}^{I2RT}_{t})/V(X_{t})$	0.917	0.142	0.071
H	$V(X_{t}-\bar{X}^{I2RT}_{t})/V(X_{t})$	0.053	1.385	0.665
	$2COV(\bar{X}^{I2RT}_{t},X_{t}-\bar{X}^{I2RT}_{t})/V(X_{t})$	0.030	-0.526	0.264
Continuing establishments only:				
	$V(\bar{X}^{I2T}_{t})/V(X_{t})$	0.802	0.098	0.044
I	$V(X_{t}-\bar{X}^{I2T}_{t})/V(X_{t})$	0.062	1.479	0.685
	$2COV(\bar{X}^{I2T}_{t},X_{t}-\bar{X}^{I2T}_{t})/V(X_{t})$	0.135	-0.577	0.272

[1]See text for explanation. Superscript definitions: I2 - 2-digit industry, I4 - 4-digit industry, R - Region, Z - Size.
[2]$V(\cdot)$ = Variance

TABLE 7a

Comovements Between Net Job Growth and Adjusted Gross Job Reallocation

Summary Statistics on Time Series Correlation of NET_{st} with:

Sector S:	SUM_{st}	$\overline{SUM}^{T}_{st}$	$\overline{SUM}^{ST}_{st}$
2-digit SIC			
Size-weighted Average Correlation	-0.51	-0.54	-0.55
# < 0/Total	20/20	20/20	20/20
Cross-sector Heterogeneity[1]	0.22	0.22	0.21
4-digit SIC			
Size-weighted Average Correlation	-0.35	-0.31	-0.36
# < 0/Total	363/450	345/450	360/450
Cross-sector Heterogeneity	0.34	0.33	0.32
Region			
Size-weighted Average Correlation	-0.51	-0.57	-0.57
# < 0/Total	9/9	9/9	9/9
Cross-sector Heterogeneity	0.23	0.15	0.18
Size			
Size-weighted Average Correlation	-0.47	-0.52	-0.53
# < 0/Total	3/3	3/3	3/3
Cross-sector Heterogeneity	0.37	0.34	0.32
2-digit, Region			
Size-weighted Average Correlation	-0.39	-0.38	-0.42
# < 0/Total	146/177	143/177	152/177
Cross-sector Heterogeneity	0.31	0.30	0.30
2-digit (continuing establishments only			
Size-weighted Average Correlation	-0.53	-0.58	-0.56
# < 0/Total	18/20	19/20	19/20
Cross-sector Heterogeneity	0.24	0.22	0.25

[1]Cross-sector heterogeneity is measured by the size-weighted standard deviation of the sectoral correlations.

TABLE 7b

Comovements Between Aggregate Net Job Growth and Adjusted Gross Job Reallocation Summary Statistics on Time Series Correlation of g_t with:

Sector S:	SUM_{st}	$\bar{SUM}^{T}_{st}$	$\bar{SUM}^{ST}_{st}$
2-digit SIC			
Size-weighted Average Correlation	-0.38	-0.48	-0.43
# < 0/Total	19/20	20/20	19/20
Cross-sector Heterogeneity	0.20	0.15	0.17
4-digit SIC			
Size-weighted Average Correlation	-0.18	-0.25	-0.19
# < 0/Total	332/450	332/450	320/450
Cross-sector Heterogeneity	0.32	0.28	0.32
Region			
Size-weighted Average Correlation	-0.44	-0.53	-0.50
# < 0/Total	8/9	9/9	8/9
Cross-sector Heterogeneity	0.28	0.20	0.24
Size			
Size-weighted Average Correlation	-0.46	-0.52	-0.52
# < 0/Total	3/3	3/3	3/3
Cross-sector Heterogeneity	0.37	0.34	0.32
2-digit, Region			
Size-weighted Average Correlation	-0.26	-0.34	-0.29
# < 0/Total	133/177	146/177	137/177
Cross-sector Heterogeneity	0.30	0.25	0.28
2-digit (continuing establishments only)			
Size-weighted Average Correlation	-0.44	-0.54	-0.47
# < 0/Total	19/20	20/20	19/20
Cross-sector Heterogeneity	0.21	0.11	0.21

[1]Cross-sector heterogeneity is measured by the size-weighted standard deviation of the sectoral correlations.

TABLE 8

Regression of Adjusted Gross Job Reallocation on Own-Sector and Manufacturing Net Growth

Dependent Variable: $\overline{SUM}^{ST}_{st}$

Regressor:	Sector S: 2-digit	4-digit	Size	Region	2-digit, Region	2-digit (Continuing Establishments Only)
g_{st}	-0.221 (0.049)[2]	-0.273 (0.029)	-0.026 (0.212)	-0.458 (0.114)	-0.296 (0.078)	-0.174 (0.052)
g_t	-0.225 (0.024)	-0.152 (0.017)	-0.238 (0.046)	-0.170 (0.026)	-0.151 (0.024)	-0.203 (0.024)
Other Summary Statistics:						
R^2 (including contribution of fixed effects)	0.78	0.53	0.91	0.63	0.29	0.75
R^2 (excluding contribution of fixed effects)	0.27	0.13	0.32	0.32	0.12	0.30
#Obs	220	4917	33	99	1936	220
Difference in coefficient on g_{st} and g_t[3]	0.004 (0.946)	-0.121 (0.0001)	0.212 (0.303)	-0.288 (0.01)	-0.145 (0.0001)	0.029 (0.448)
Mean ($\overline{SUM}^{ST}_{st}$)	0.202	0.209	0.195	0.210	0.216	0.156
Std. Dev. ($\overline{SUM}^{ST}_{st}$)	0.049	0.096	0.059	0.029	0.081	0.033
Mean (g_t)	-0.021	-0.021	-0.021	-0.021	-0.021	-0.021
Std. Dev. (g_t)	0.052	0.052	0.052	0.052	0.052	0.046
Mean (g_{st})	-0.003	-0.009	0.0002	0.006	0.0001	-0.001
Std. Dev. (g_{st})	0.042	0.091	0.011	0.023	0.075	0.039

[1]All regressions include sector fixed effects.
[2]Heteroscedasticity-consistent standard errors in parentheses (using White correction).
[3]Marginal significance level on Chi-square test that coefficients are equal.

FIGURE 1A

SIZE-WEIGHTED GROWTH RATE DISTRIBUTION

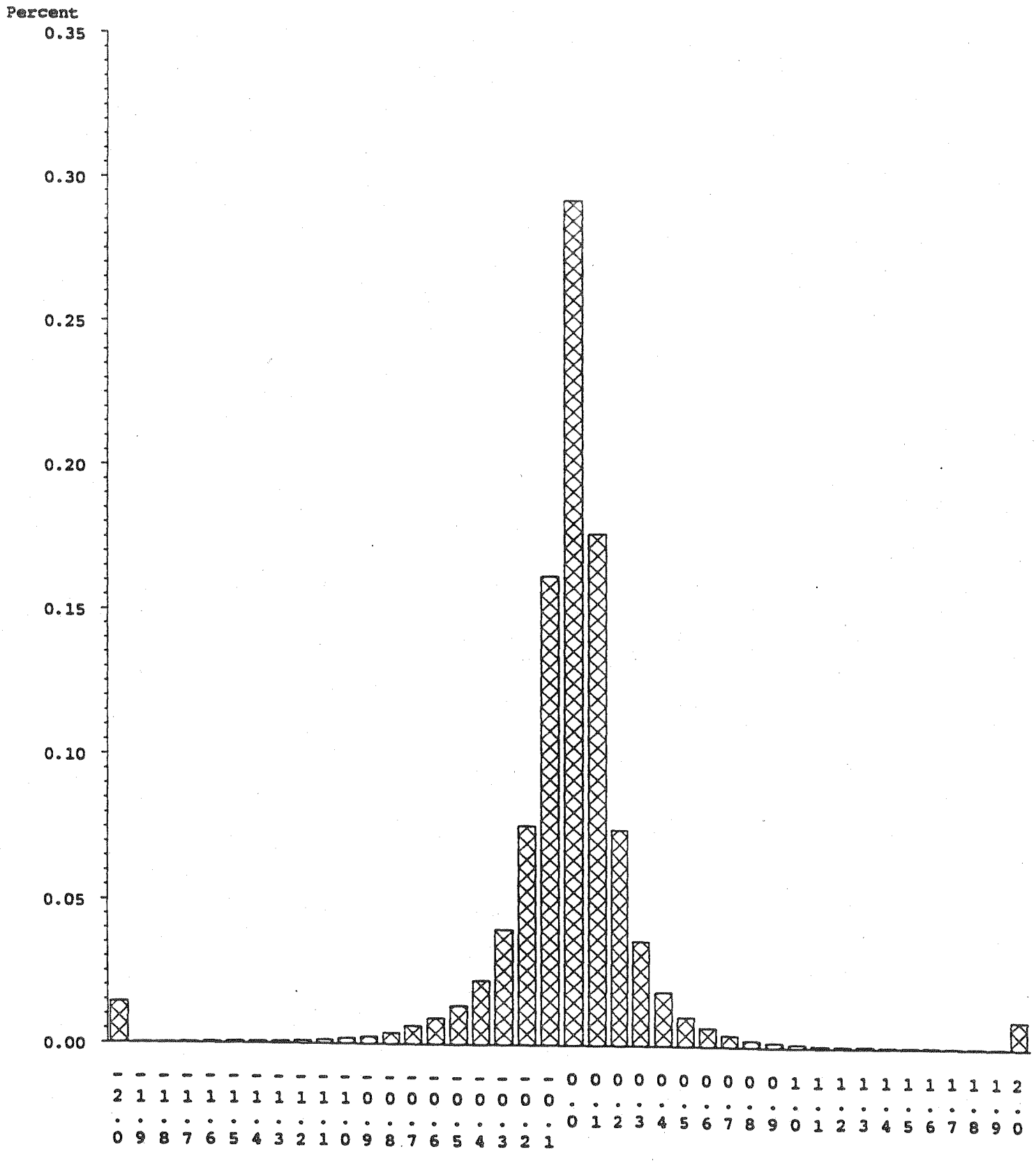


FIGURE 1B

GROWTH RATE DISTRIBUTION

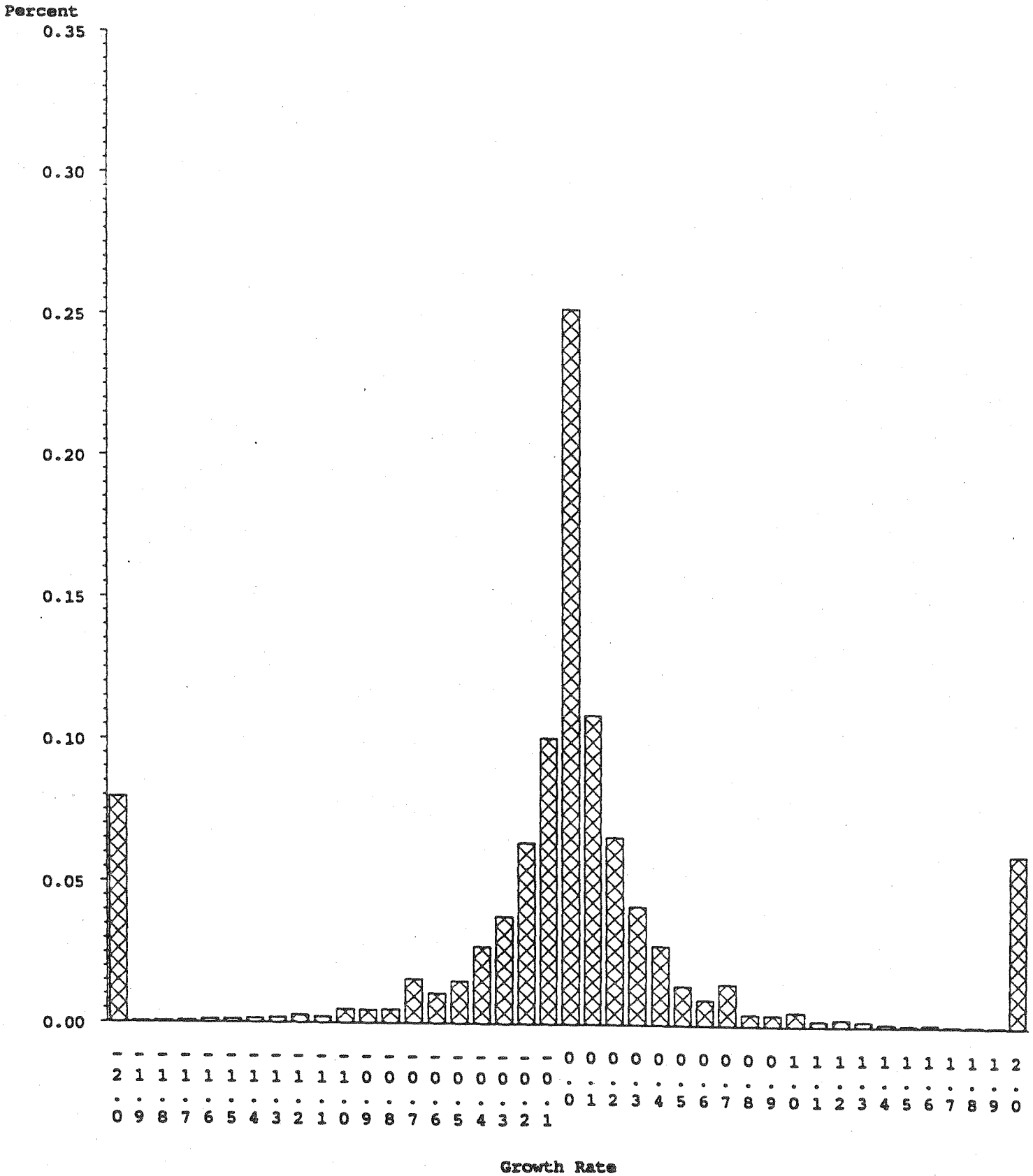
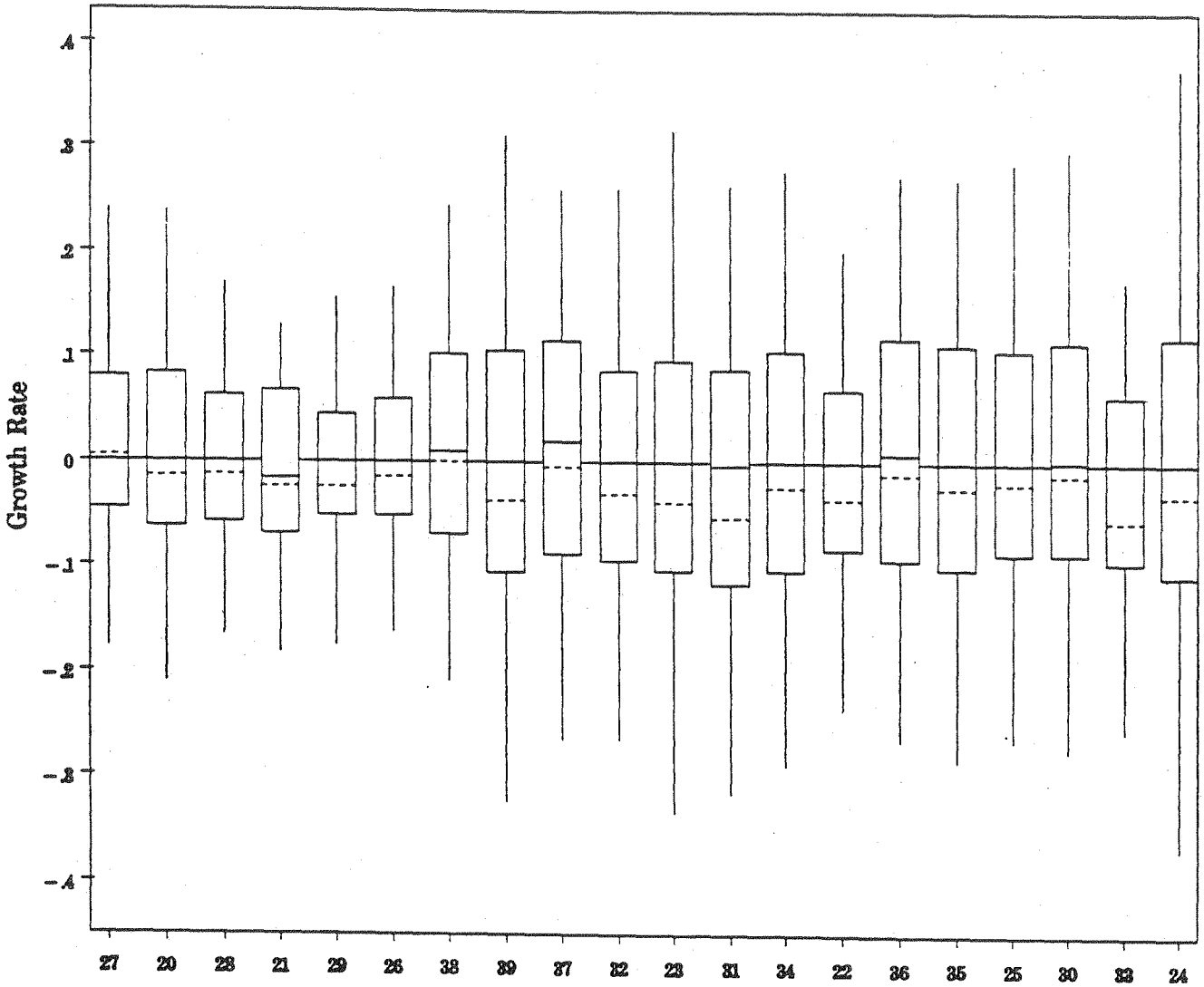


Figure 2

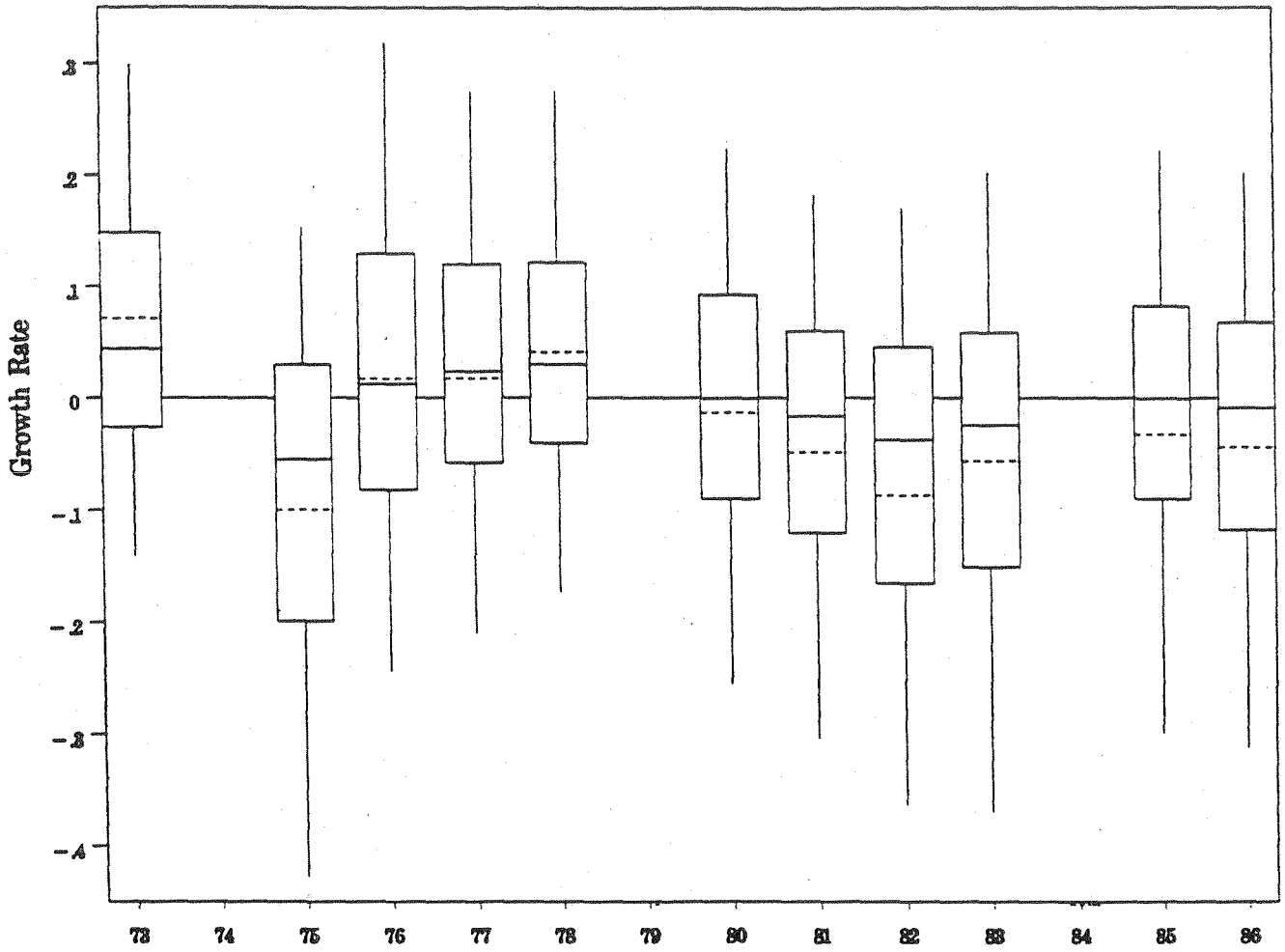
Box Plots for Industry Growth Rate Distributions



NOTE: The box plots show the 10th, 25th, 50th, 75th, and 90th percentiles. The dashed line is the mean.

Figure 3

Box Plots for Annual Growth Rate Distributions



NOTE: The box plots show the 10th, 25th, 50th, 75th, and 90th percentiles. The dashed line is the mean.